

Running Jobs: Part 1: Batch Computing Part 2: Interactive Computing

COMPLECS Team

<https://bit.ly/COMPLECS>

<https://github.com/sdsc-complecs>

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COMPLECS: Batch Computing Series

Working with the Linux Scheduler

How to interact with the Linux scheduler and run your research workloads on your personal computer, a shared workstation, or even a high-performance computing system.

Getting Started with Batch Job Scheduling: Slurm Edition

How to schedule your batch jobs on high-performance computing systems using the Slurm Workload Manager.

High-Throughput and Many-Task Computing Workflows: Slurm Edition

How to build and run your high-throughput and many-task computing workflows on high-performance computing systems using the Slurm Workload Manager.





Getting Started with Batch Job Scheduling: Slurm Edition

Learning Goals:

- What is a batch job?
- What is Slurm?
- How to start interacting with Slurm
- How to write, submit and run your first batch job on a high-performance computing system

Prerequisite Knowledge:

- [Parallel Computing Concepts](#)
- [Intermediate Linux and Shell Scripting](#)

Interactive High-performance (HPC) Computing

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<https://bit.ly/COMPLECS>

<https://github.com/sdsc-complecs/interactive-computing/>

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Table of Contents: Part 1

- **High-Performance Computing Hardware and System Architecture**
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 - **squeue** - view information about jobs located in the Slurm scheduling queue
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 - **sacct** - displays accounting data for all jobs in the Slurm database
- **Best Practices with Slurm**

HPC Compute Node: Dell C4140



Figure 1. Front view of the system

1. Control panel



Figure 2. Rear view of the system

1. PCIe expansion card slot 1
 2. PCIe expansion card slot 2
 3. PCIe expansion card slot 3
 4. Power supply unit (PSU 1)
 5. Power supply unit (PSU 2)
 6. Ethernet connectors (4)
 7. USB connectors (2)
 8. Video connector
 9. Serial connector
 10. iDRAC Enterprise port
 11. NMI button
 12. System identification button
- NOTE:** This slot is dedicated for BOSS c

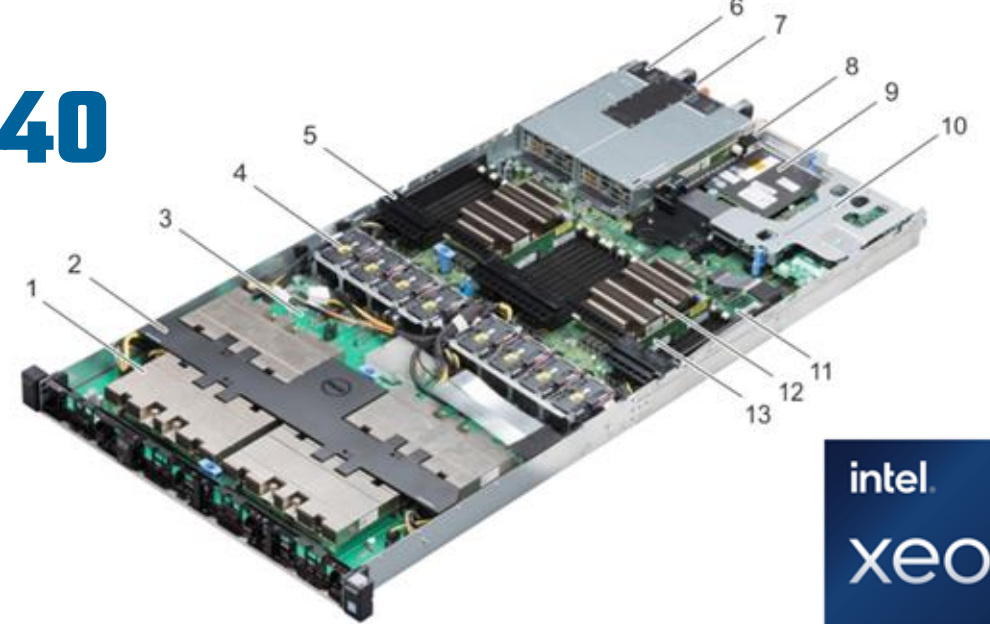


Figure 4. Configuration K

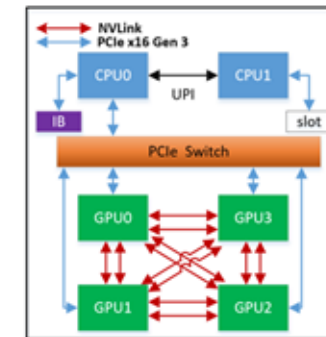
1. NVLink Heat sink and processor (4)
2. NVLink air shroud
3. NVLink board
4. Cooling fan (8)
5. Air shroud
6. PSU (2)
7. Information tag
8. Riser 2A (Low-profile PCIe expansion card: Slot 3)
9. Network daughter card (NDC)
10. Riser 1A (Low-profile PCIe expansion cards: Slots 1 and 2)
11. System board
12. Processor and heat sink (2)
13. DIMMs (24)



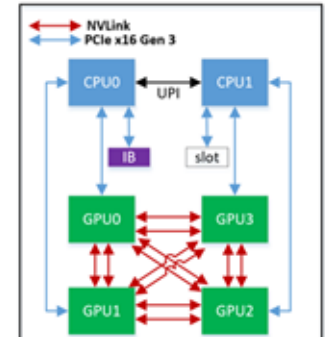
[Intel Xeon Gold 6248](#)



[NVIDIA V100 32GB SMX2](#)



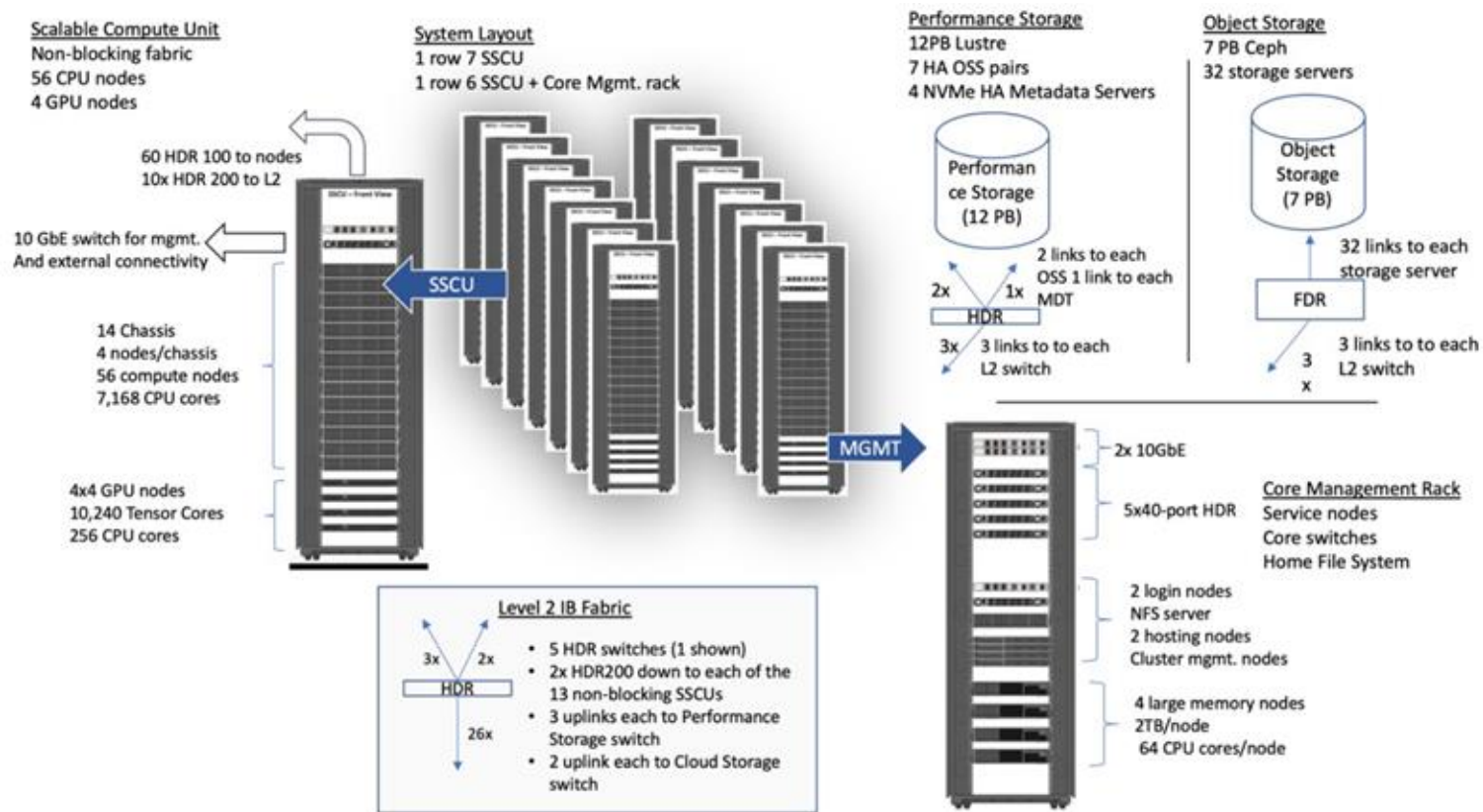
(a) Configuration K



(b) Configuration M

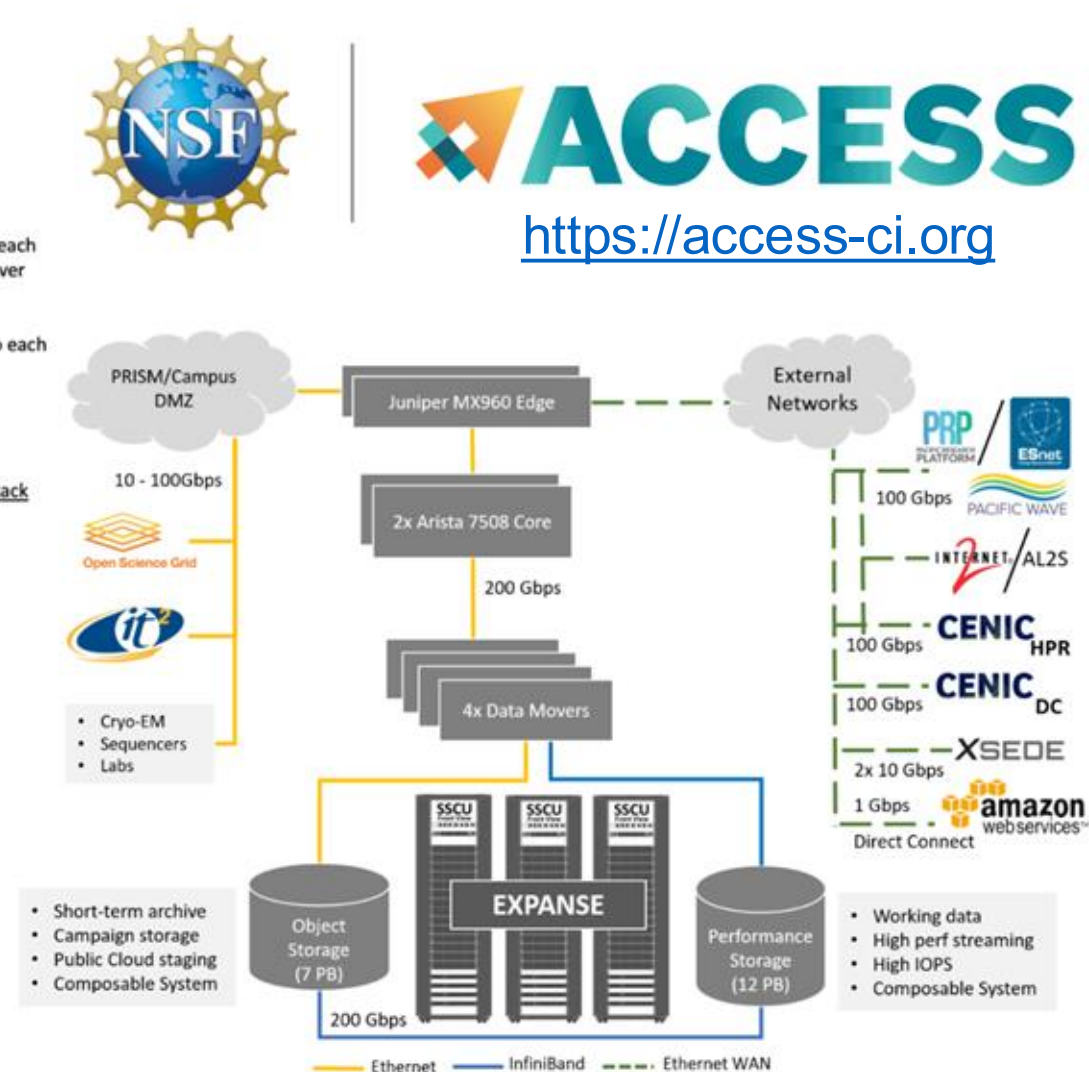
Figure 1: The comparison between configuration K and configuration M in C4140 server

HPC System Architecture: Expanse @ SDSC



<https://expanse.sdsc.edu>

https://www.youtube.com/watch?v=uNZyg6X_t3s



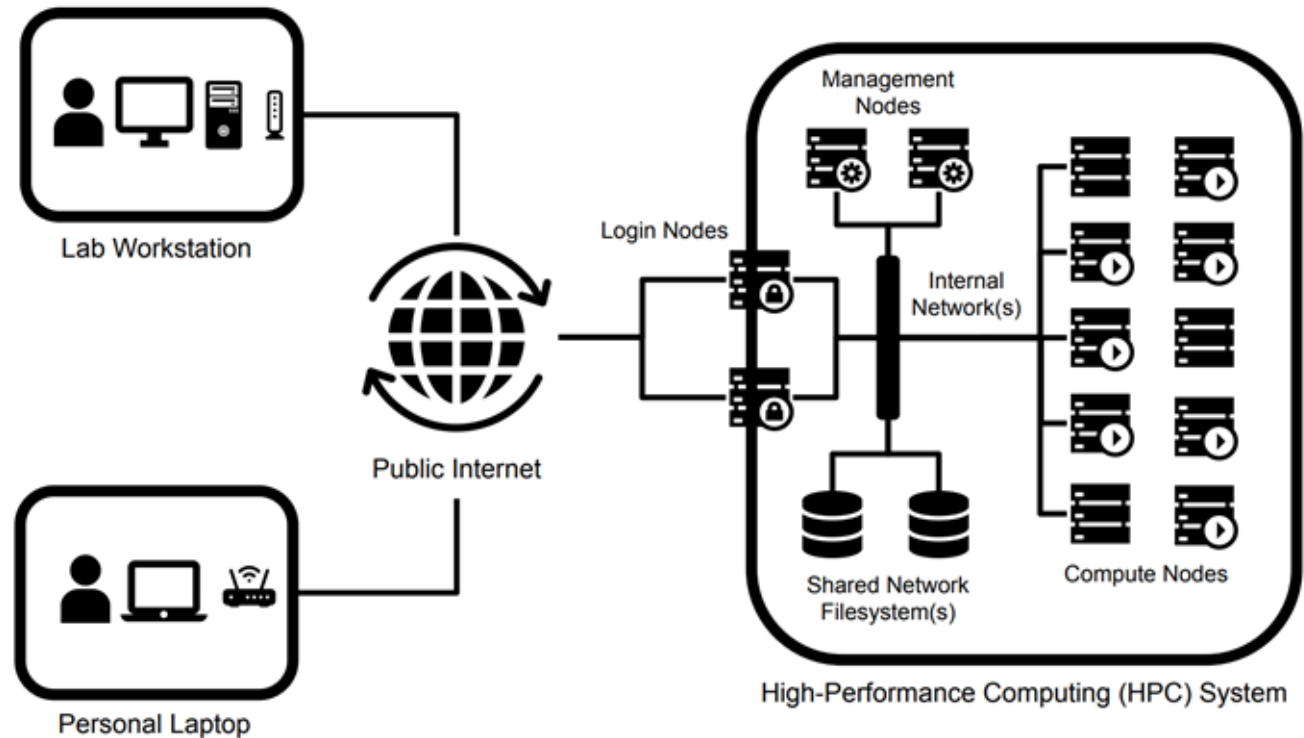
HPC System Architecture: Conceptual Model

Login node(s): Provide remote access to an HPC system; use only for simple tasks such as editing files, limited data transfers to and from the system, and batch job submission

Compute nodes: Run computational workloads: simulations, data analysis and visualization

Internal Network(s): Provide high-bandwidth, low-latency communication between compute nodes ; access to shared (parallel) filesystems; system management

Shared Network Filesystem(s): Provide input/output (I/O) access to data storage systems from any compute node



Management node(s): Run core system services such as cluster management software, system monitoring software, *batch job scheduler*, etc

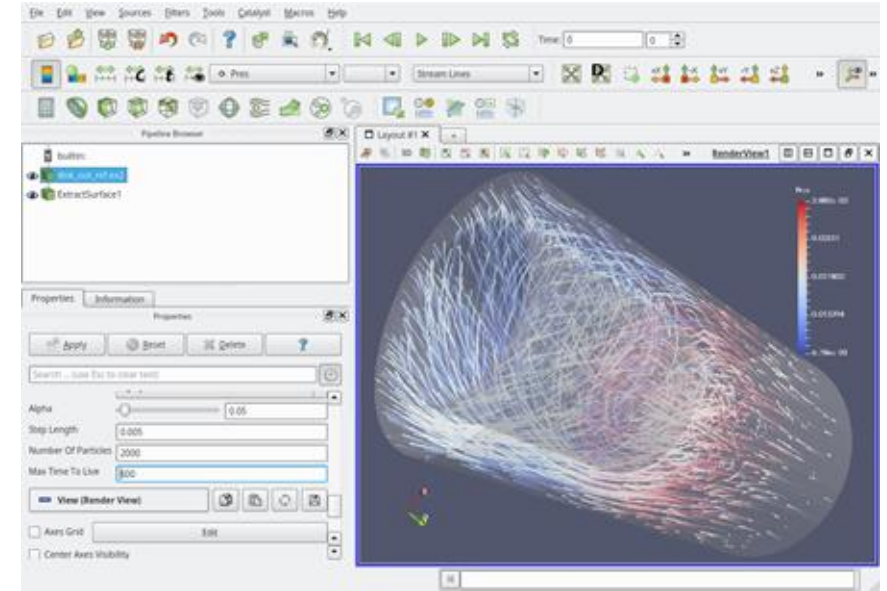
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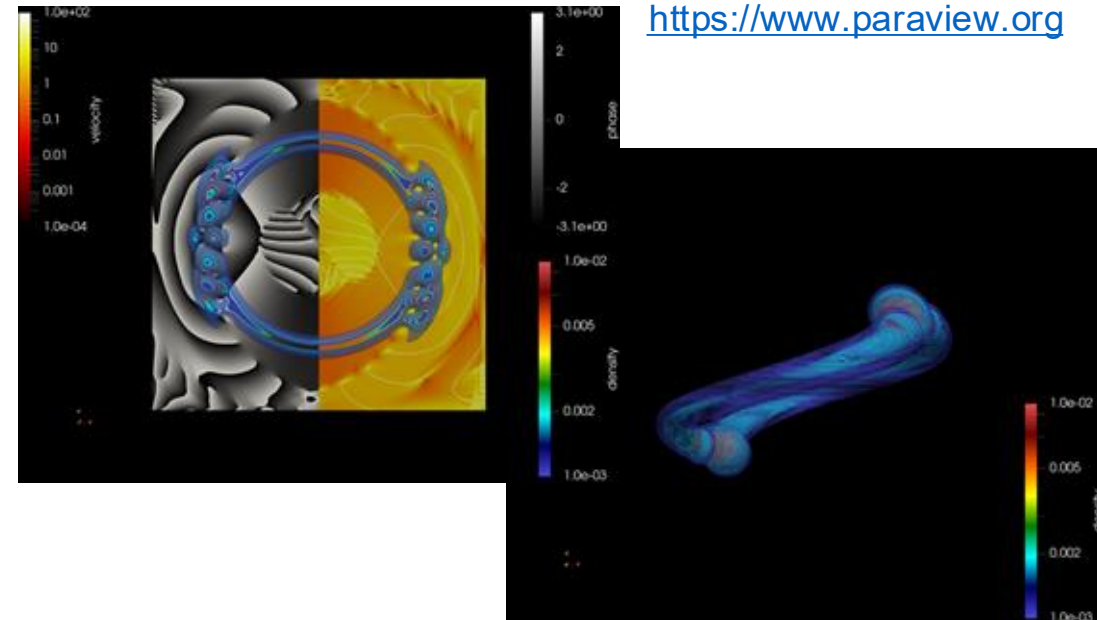
Interactive vs. Batch Computing

Interactive computing is the use of computer programs that accept input from the user as they execute on a system. Examples in HPC include the use of software applications for code development and performance optimization as well as real-time data exploration, analysis and visualization.

Batch processing involves the execution of computer programs in an automated and unattended way on a system without user interaction. Batch computing systems are widely used to automate high-volume, repetitive tasks such as data processing and to ***allow the shared use of advanced compute resources like an HPC system*** among many users.



<https://www.paraview.org>

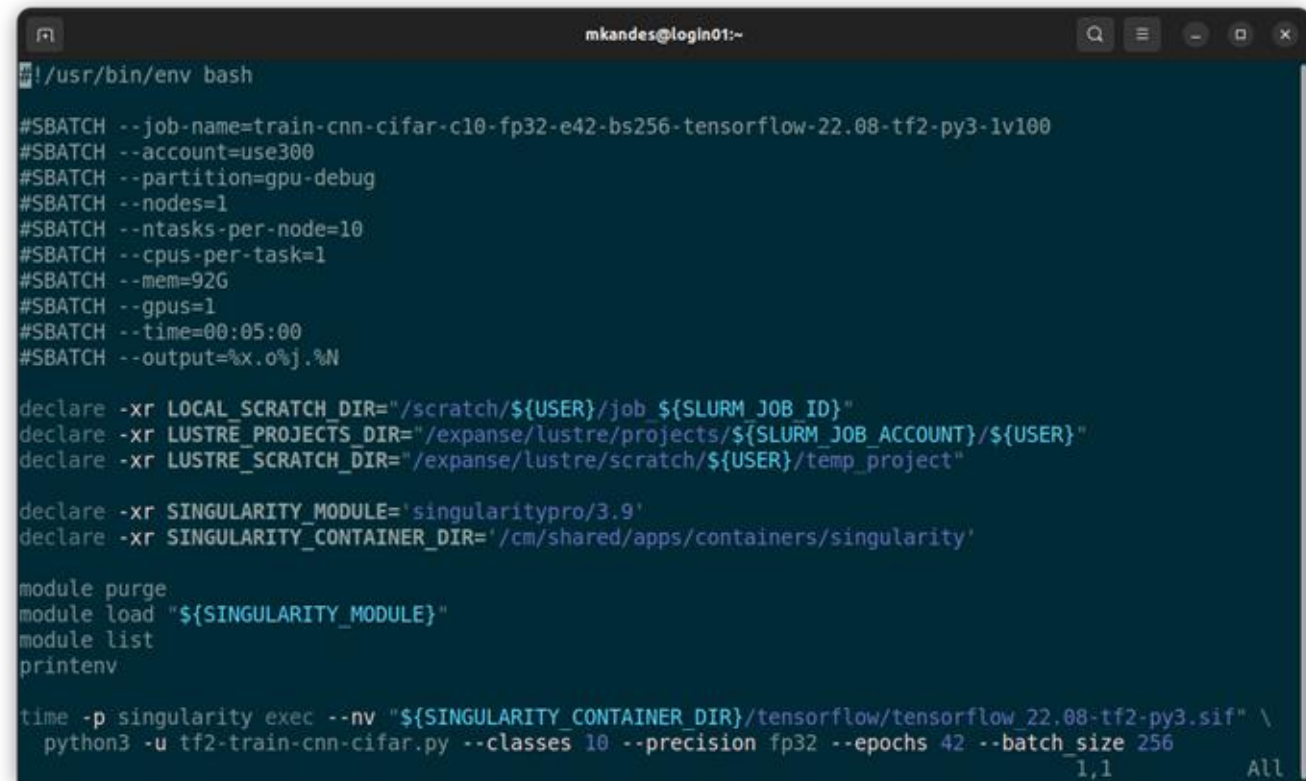


What is a Batch Job?

A **batch job** is a *pre-scripted* sets of commands that are executed as processes, which may or may not be multi-threaded, on a certain type or set of dedicated compute resources within a computing system for a given amount of time without user interaction.

On HPC systems, batch jobs may run large-scale simulations, perform complex data analysis, or any other types of demanding computational tasks.

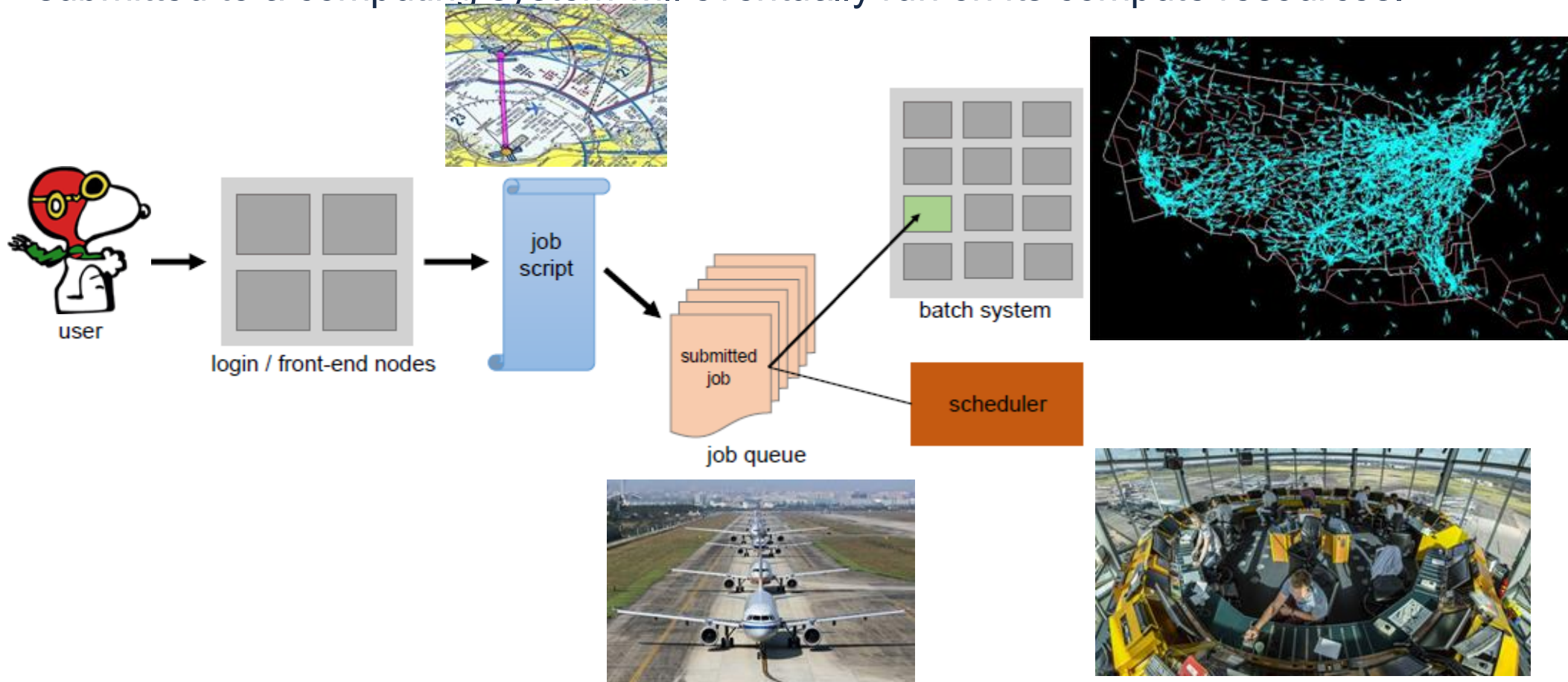
A batch job is specified as a standard shell script that is then submitted to a queue of other batch jobs managed by a scheduler, which assigns the job to available resources and ensures that the job runs to completion.



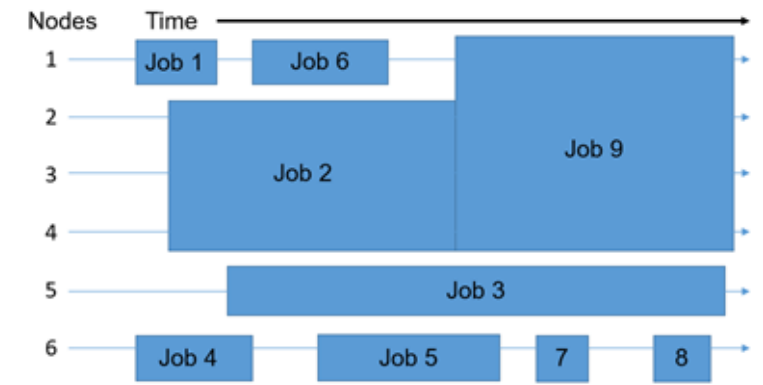
```
mkandes@login01:~  
#!/usr/bin/env bash  
  
#SBATCH --job-name=train-cnn-cifar-c10-fp32-e42-bs256-tensorflow-22.08-tf2-py3-1v100  
#SBATCH --account=use300  
#SBATCH --partition=gpu-debug  
#SBATCH --nodes=1  
#SBATCH --ntasks-per-node=10  
#SBATCH --cpus-per-task=1  
#SBATCH --mem=92G  
#SBATCH --gpus=1  
#SBATCH --time=00:05:00  
#SBATCH --output=%x.0%j.%N  
  
declare -xr LOCAL_SCRATCH_DIR="/scratch/${USER}/job_${SLURM_JOB_ID}"  
declare -xr LUSTRE_PROJECTS_DIR="/exppanse/lustre/projects/${SLURM_JOB_ACCOUNT}/${USER}"  
declare -xr LUSTRE_SCRATCH_DIR="/exppanse/lustre/scratch/${USER}/temp_project"  
  
declare -xr SINGULARITY_MODULE='singularitypro/3.9'  
declare -xr SINGULARITY_CONTAINER_DIR="/cm/shared/apps/containers/singularity"  
  
module purge  
module load "${SINGULARITY_MODULE}"  
module list  
printenv  
  
time -p singularity exec --nv "${SINGULARITY_CONTAINER_DIR}/tensorflow/tensorflow_22.08-tf2-py3.sif" \  
python3 -u tf2-train-cnn-cifar.py --classes 10 --precision fp32 --epochs 42 --batch_size 256  
1,1 All
```

What is a Batch Job Scheduler?

A **batch job scheduler** is software that controls and tracks where and when batch jobs submitted to a computing system will eventually run on its compute resources.



Batch Job Scheduling in Practice



Batch job schedulers typically have three primary aims:

- Minimize the time between the job submission and job completion: no user job should stay pending in the queue for too long
- Optimize compute resource utilization: no compute resource should be idle for too long
- Maximize job throughput: process as many jobs as soon as possible

In most cases, the batch job schedulers on HPC systems will employ some form of a [fairshare scheduling](#) algorithm to match user jobs to compute nodes over time.

** All interactive jobs you run on an HPC system are usually treated the same as batch jobs.*

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Slurm Workload Manager

The [Slurm Workload Manager](https://slurm.schedmd.com) is a free and open-source distributed batch job scheduler and resource manager for Unix-like operating systems used by the majority of the world's supercomputers and computer clusters today.

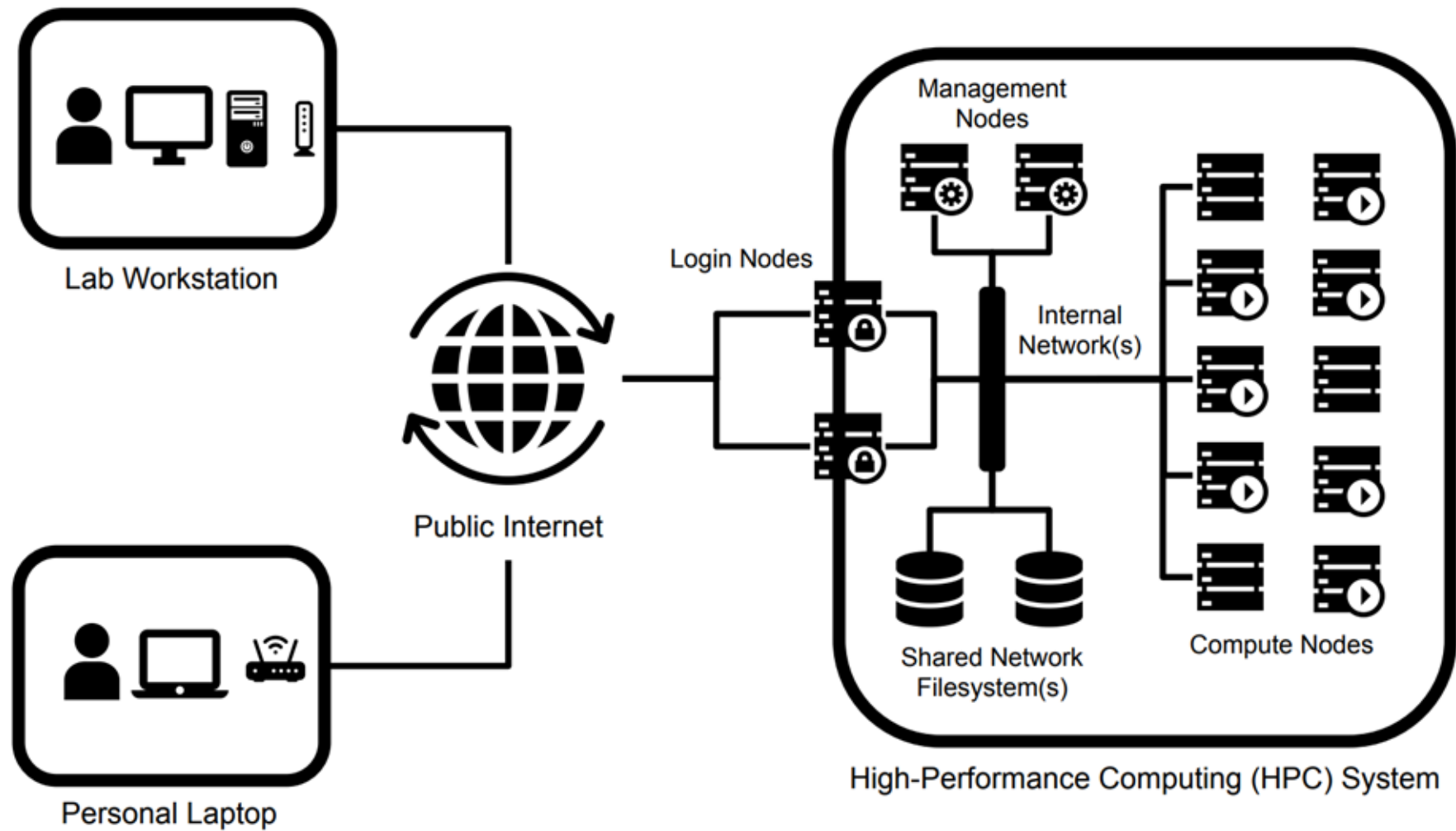
Slurm provides three key functions:

- the allocation of compute nodes for some period of time to process user jobs;
- a framework for starting, executing, and monitoring the progress of user jobs across a set of allocated compute nodes; and
- arbitrating the demand for compute nodes by managing a queue of pending user jobs

For more information, see the Slurm documentation: <https://slurm.schedmd.com>



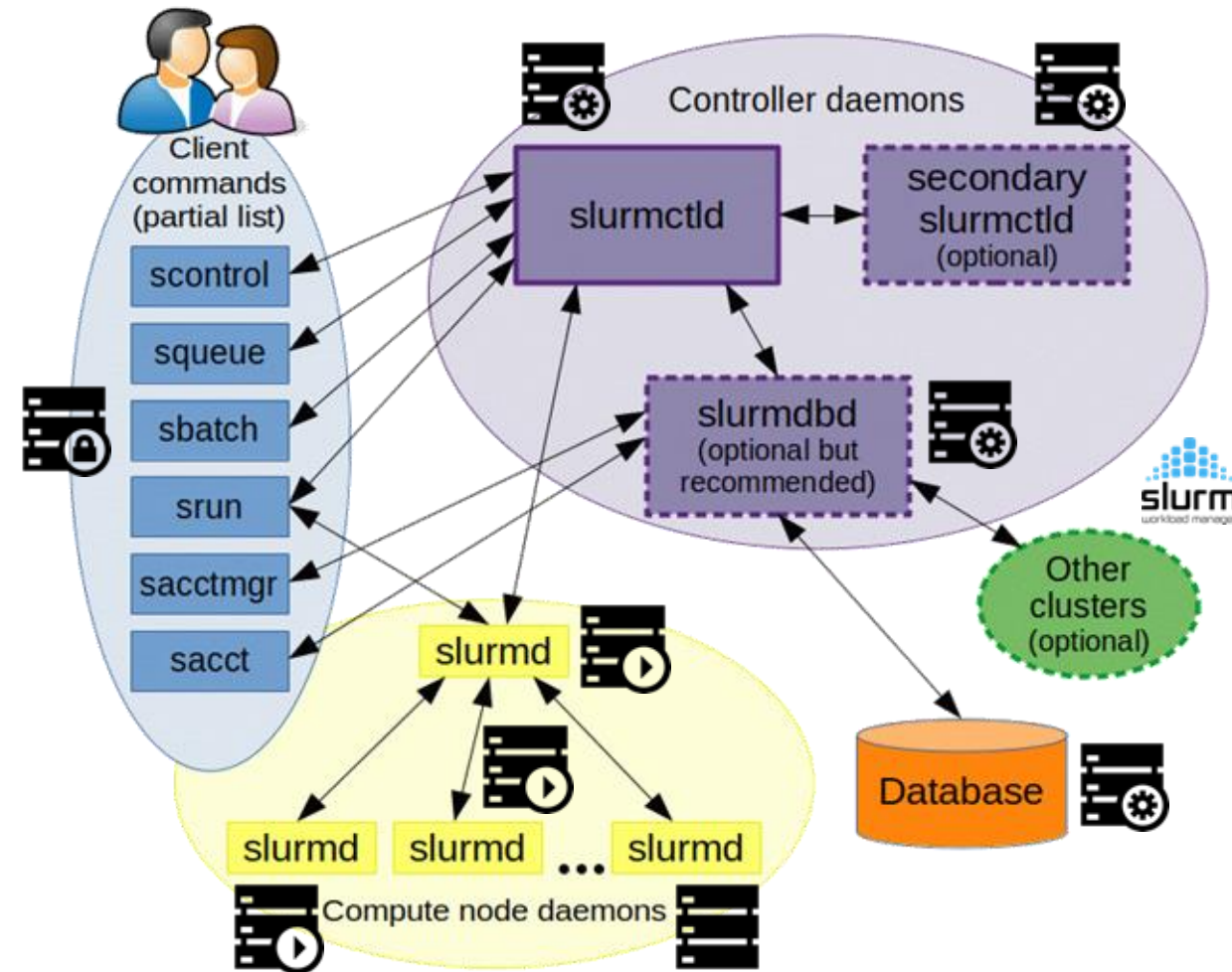
HPC System Architecture: Conceptual Model



Slurm Architecture

Slurm is comprised of the three following types of service daemons:

- **slurmd** is the compute node daemon; it monitors all jobs running on the compute node, accepts jobs, launches jobs, and stops running jobs
- **slurmctld** is the central management daemon; it monitors all other Slurm daemons and resources, accepts jobs, and allocates resources to the jobs.
- **slurmdbd** is the accounting database daemon; it provides an interface for archiving accounting records to an external database



<https://slurm.schedmd.com/quickstart.html>

Slurm User Commands

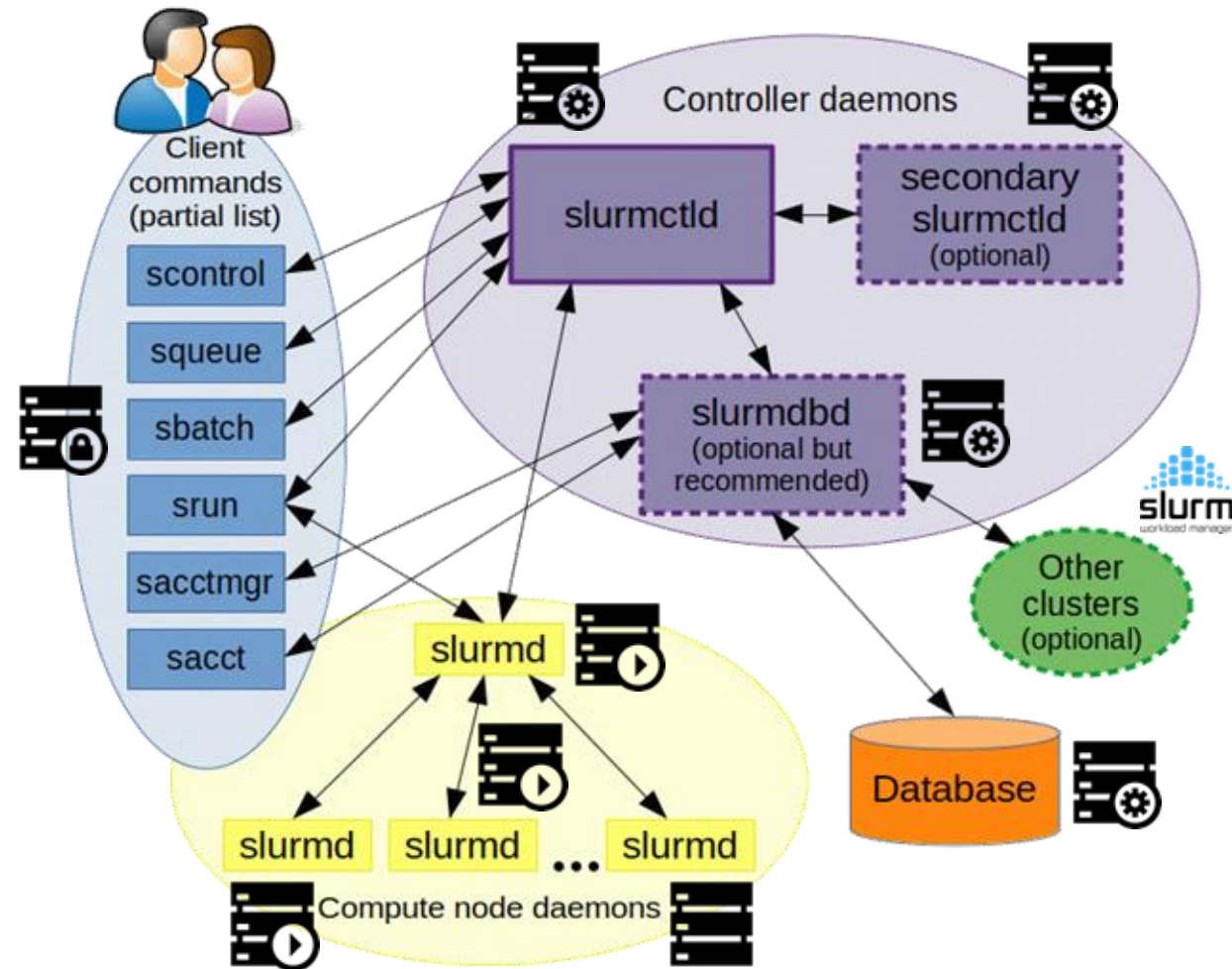
The essential client commands you'll want to learn how to use as you get started with Slurm are as follows:

- [sbatch](#)
- [squeue](#)
- [scancel](#)

Other client commands you may find useful from time to time are:

- [sacct](#)
- [sinfo](#)

You will generally only run these commands interactively on a system's login nodes.



<https://slurm.schedmd.com/quickstart.html>

squeue

The `squeue` command is used to query and view information about the user jobs submitted to the Slurm scheduling queue.

```
mkandes@login01:~$ squeue
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
28825741	compute	touch	fzzhao	PD	0:00	1	(MaxMemPerLimit)
29153659	compute	P353	dc1993	PD	0:00	1	(MaxMemPerLimit)
29153657	compute	P353	dc1993	PD	0:00	1	(MaxMemPerLimit)
29184587	compute	V0223_CT	avibho	PD	0:00	2	(MaxMemPerLimit)
28970190	compute	Rxn7	saleh35	PD	0:00	5	(MaxMemPerLimit)
28970185	compute	Rxn5	saleh35	PD	0:00	5	(MaxMemPerLimit)
28970184	compute	Rxn4	saleh35	PD	0:00	5	(MaxMemPerLimit)
28970183	compute	Mo-Pd	saleh35	PD	0:00	5	(MaxMemPerLimit)
28970181	compute	Rxn2	saleh35	PD	0:00	5	(MaxMemPerLimit)
29300918	compute	V1.sim	chenziao	PD	0:00	1	(MaxMemPerLimit)
29372691	compute	18-10-12	vdonmes	PD	0:00	10	(Resources)
29372688	compute	18-6-128	vdonmes	PD	0:00	6	(Priority)
29372690	compute	18-6-128	vdonmes	PD	0:00	6	(Priority)
29372756	compute	ceMvp-0	aikeliu	PD	0:00	5	(Priority)
29373676	compute	bs.fe	prakashl	PD	0:00	2	(Priority)
29374534	compute	A6007220	us3	PD	0:00	1	(Priority)
29373711	compute	bs.fe	prakashl	PD	0:00	2	(Priority)
29373974	compute	A3aa	oohiro	PD	0:00	2	(Priority)
29373975	compute	A3ab	oohiro	PD	0:00	2	(Priority)
29374059	compute	A3ba	oohiro	PD	0:00	2	(Priority)
29374061	compute	A3bb	oohiro	PD	0:00	2	(Priority)
29374083	compute	A1c	oohiro	PD	0:00	2	(Priority)
29374940	compute	1.15.csh	amy44ric	PD	0:00	1	(Priority)
29374717	compute	n32-t1-m	jackyanl	PD	0:00	32	(Priority)
29374715	compute	n32-t1-l	jackyanl	PD	0:00	32	(Priority)
29374716	compute	n32-t1-m	jackyanl	PD	0:00	32	(Priority)
29374727	compute	n32-t1-l	jackyanl	PD	0:00	32	(Priority)
29374546	compute	SHIRLEY	sjammuch	PD	0:00	2	(Priority)
29373790	compute	CH0_CHfs	yuting82	PD	0:00	2	(Priority)
29374600	compute	cp2k-com	wborrell	PD	0:00	1	(Priority)
29374602	compute	cp2k-com	wborrell	PD	0:00	1	(Priority)
29375307	compute	1.15.csh	amy44ric	PD	0:00	1	(Priority)
29375309	compute	14.1.15.	amy44ric	PD	0:00	1	(Priority)
29374044	compute	vasp6-co	zwang21	PD	0:00	3	(Priority)

In general, you should only need to check the status of the jobs you've submitted to Slurm.

Recommendation: Create an alias in your shell's configuration file to only check the status of your jobs.

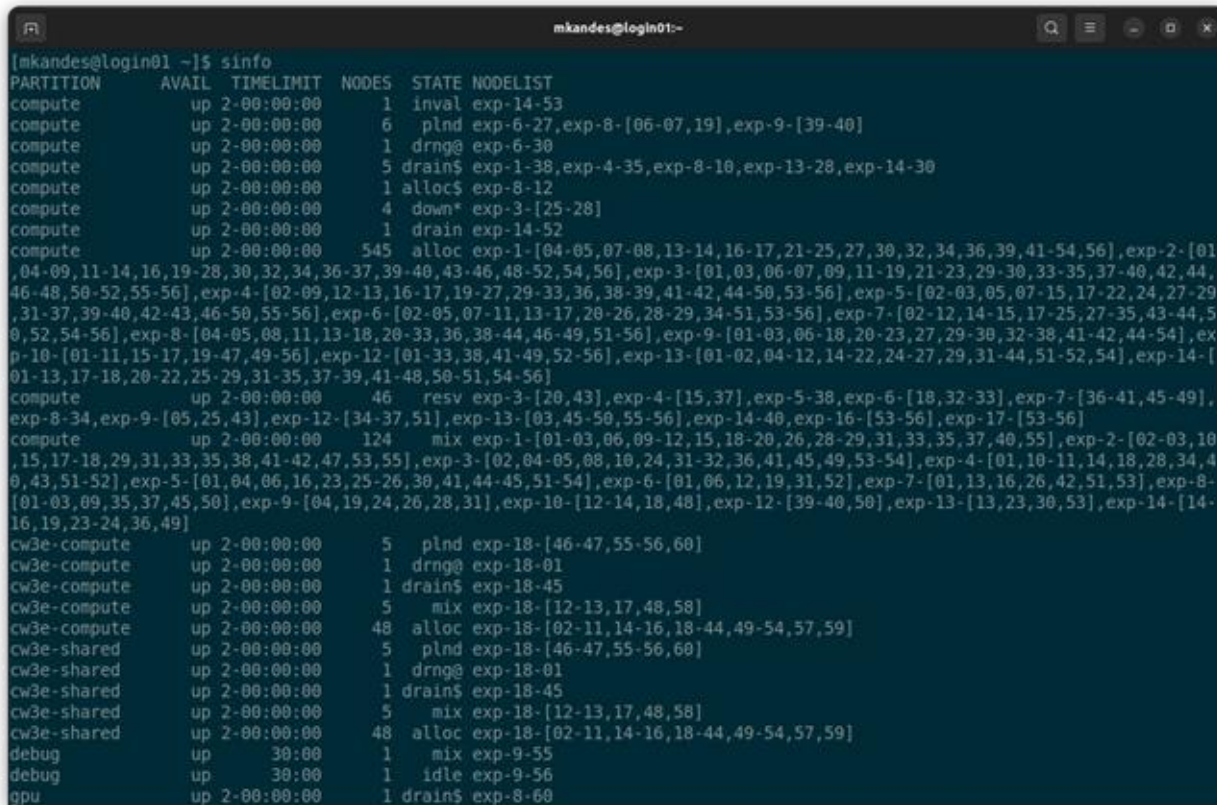
```
alias squeue="squeue -u ${USER}"
```

Do **NOT** run the `squeue` command incessantly to poll job status, either on purpose or on accident.



sinfo

The [sinfo](#) command is used to query and view information about the compute nodes and partitions managed by Slurm.



```
mkandes@login01:~$ sinfo
PARTITION  AVAIL  TIMELIMIT  NODES  STATE  NODELIST
compute    up 2-00:00:00      1  inval exp-14-53
compute    up 2-00:00:00      6  plnd  exp-6-27,exp-8-[06-07,19],exp-9-[39-40]
compute    up 2-00:00:00      1  drng@ exp-6-30
compute    up 2-00:00:00      5  drain$ exp-1-38,exp-4-35,exp-8-10,exp-13-28,exp-14-30
compute    up 2-00:00:00      1  alloc$ exp-8-12
compute    up 2-00:00:00      4  down*  exp-3-[25-28]
compute    up 2-00:00:00      1  drain exp-14-52
compute    up 2-00:00:00     545  alloc exp-1-[04-05,07-08,13-14,16-17,21-25,27,30,32,34,36,39,41-54,56],exp-2-[01-04,09,11-14,16,19-28,30,32,34,36-37,39-40,43-46,48-52,54,56],exp-3-[01,03,06-07,09,11-19,21-23,29-30,33-35,37-40,42,44,46-48,50-52,55-56],exp-4-[02-09,12-13,16-17,19-27,29-33,36,38-39,41-42,44-50,53-56],exp-5-[02-03,05,07-15,17-22,24,27-29,31-37,39-40,42-43,46-50,55-56],exp-6-[02-05,07-11,13-17,20-26,28-29,34-51,53-56],exp-7-[02-12,14-15,17-25,27-35,43-44,50,52,54-56],exp-8-[04-05,08,11,13-18,20-33,36,38-44,46-49,51-56],exp-9-[01-03,06-18,20-23,27,29-30,32-38,41-42,44-54],exp-10-[01-11,15-17,19-47,49-56],exp-12-[01-33,38,41-49,52-56],exp-13-[01-02,04-12,14-22,24-27,29,31-44,51-52,54],exp-14-[01-13,17-18,20-22,25-29,31-35,37-39,41-48,50-51,54-56]
compute    up 2-00:00:00      46  resv  exp-3-[20,43],exp-4-[15,37],exp-5-38,exp-6-[18,32-33],exp-7-[36-41,45-49],exp-8-34,exp-9-[05,25,43],exp-12-[34-37,51],exp-13-[03,45-50,55-56],exp-14-40,exp-16-[53-56],exp-17-[53-56]
compute    up 2-00:00:00     124  mix   exp-1-[01-03,06,09-12,15,18-20,26,28-29,31,33,35,37,40,55],exp-2-[02-03,10,15,17-18,29,31,33,35,38,41-42,47,53,55],exp-3-[02,04-05,08,10,24,31-32,36,41,45,49,53-54],exp-4-[01,10-11,14,18,28,34,40,43,51-52],exp-5-[01,04,06,16,23,25-26,30,41,44-45,51-54],exp-6-[01,06,12,19,31,52],exp-7-[01,13,16,26,42,51,53],exp-8-[01-03,09,35,37,45,50],exp-9-[04,19,24,26,28,31],exp-10-[12-14,18,48],exp-12-[39-40,50],exp-13-[13,23,30,53],exp-14-[14-16,19,23-24,36,49]
cw3e-compute    up 2-00:00:00      5  plnd  exp-18-[46-47,55-56,60]
cw3e-compute    up 2-00:00:00      1  drng@ exp-18-01
cw3e-compute    up 2-00:00:00      1  drain$ exp-18-45
cw3e-compute    up 2-00:00:00      5  mix   exp-18-[12-13,17,48,58]
cw3e-compute    up 2-00:00:00     48  alloc exp-18-[02-11,14-16,18-44,49-54,57,59]
cw3e-shared     up 2-00:00:00      5  plnd  exp-18-[46-47,55-56,60]
cw3e-shared     up 2-00:00:00      1  drng@ exp-18-01
cw3e-shared     up 2-00:00:00      1  drain$ exp-18-45
cw3e-shared     up 2-00:00:00      5  mix   exp-18-[12-13,17,48,58]
cw3e-shared     up 2-00:00:00     48  alloc exp-18-[02-11,14-16,18-44,49-54,57,59]
debug          up      30:00      1  mix   exp-9-55
debug          up      30:00      1  idle  exp-9-56
gpu            up 2-00:00:00      1  drain$ exp-8-60
```

Slurm partitions are used to group nodes into logical, sometimes overlapping sets of nodes.

Each partition may have a unique set of constraints and features such as job size and runtime limits, types of resources available, user-based access permissions, etc.

You should always select the most appropriate partition for any job.

https://www.sdsc.edu/support/user_guides/expanse.html#running

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Batch Scripts Contents

```
[username@login02 calc-prime]$ cat mpi-prime-slurm.sb
#!/bin/bash
#SBATCH --job-name="mpi_prime"
#SBATCH --output="mpi_prime.%j.%N.out"
####SBATCH --partition=compute
#SBATCH --partition=debug
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --export=ALL
#SBATCH -t 00:10:00
#SBATCH -A use300

## Environment
module purge
module load slurm
module load cpu
module load gcc/10.2.0
module load openmpi/4.0.4

## echo job name and id:
echo "SLURM_JOB_NAME: $SLURM_JOB_NAME"
echo "SLURM_JOB_ID: $SLURM_JOB_ID"
d=`date`
echo "DATE: $d"
```

Batch Script 4 Main Parts:

- “Shebang” !
 - `#!/bin/bash` command tells the shell to run the script using the bash
- Resource Request:
 - options preceded with `"#SBATCH"` before any executable commands in the script
- Dependencies
 - Loads software that the project depends on to execute
- Job Steps:
 - Specify the list of tasks to be carried out.

Batch Script Output: ENV_Info

```
[username@login02 env_info]$ cat env-slurm.sb
#!/bin/bash
#SBATCH --job-name=" env_info "
#SBATCH --output="mpi_prime.%j.%N.out"
####SBATCH --partition=compute
#SBATCH --partition=debug
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --export=ALL
#SBATCH -t 00:10:00
#SBATCH -A use300
```

```
## Environment
module purge
module load slurm
module load cpu
module load gcc/10.2.0
module load openmpi/4.0.4
```

```
## echo job name and id:
echo "SLURM_JOB_NAME: $SLURM_JOB_NAME"
echo "SLURM_JOB_ID: $SLURM_JOB_ID"
d=`date`
echo "DATE: $d"
e=`env`
echo "env= $e"
```

```
[mthomas@login02 env_info]$ sbatch env-slurm.sb
Submitted batch job 14126259
[mthomas@login02 env_info]$ lsq
queue -u mthomas
JOBID PARTITION  NAME  USER ST  TIME  NODES NODELIST(REASON)
14126259  debug env_info  mthomas R   0:04   1 exp-9-55
[mthomas@login01 env_info]$ cat env_info.14126259.exp-4-35.out
SLURM_JOB_NAME: env_info
SLURM_JOB_ID: 14126259
hostname= exp-4-35
date= Sun Jun 26 22:05:15 PDT 2022
whoami= mthomas
pwd= /home/mthomas/hpctr-examples/env_info
Currently Loaded Modules: 1) slurm/expanse/21.08.8 2) cpu/0.15.4
-----
env= LD_LIBRARY_PATH=/cm/shared/apps/slurm/current/lib64/slurm:
[SNIP]
/cm/shared/apps/slurm/current/lib64
SLURM_SUBMIT_DIR=/home/mthomas/hpctr-examples/env_info
HISTCONTROL=ignoredups
DISPLAY=localhost:16.0
HOSTNAME=exp-4-35
[SNIP]
```


General Steps: Compiling/Running Jobs

- Change to your working directory (e.g. hpctr-examples):

```
[mthomas@login02 ~]$  
[mthomas@login02 ~]$ cd  
/home/$USER/hpctr-examples/mpi  
[mthomas@login02 mpi]$ ll hello_mpi.f90  
-rw-r--r-- 1 mthomas use300 333 Feb 18 13:42  
hello_mpi.f90  
[mthomas@login02 mpi]$
```

```
[mthomas@login02 ~]$ module list  
Currently Loaded Modules:  
  1) shared  2) cpu/0.15.4  3)  
slurm/expanse/21.08.8  
  4) sdsc/1.0  5) DefaultModules
```

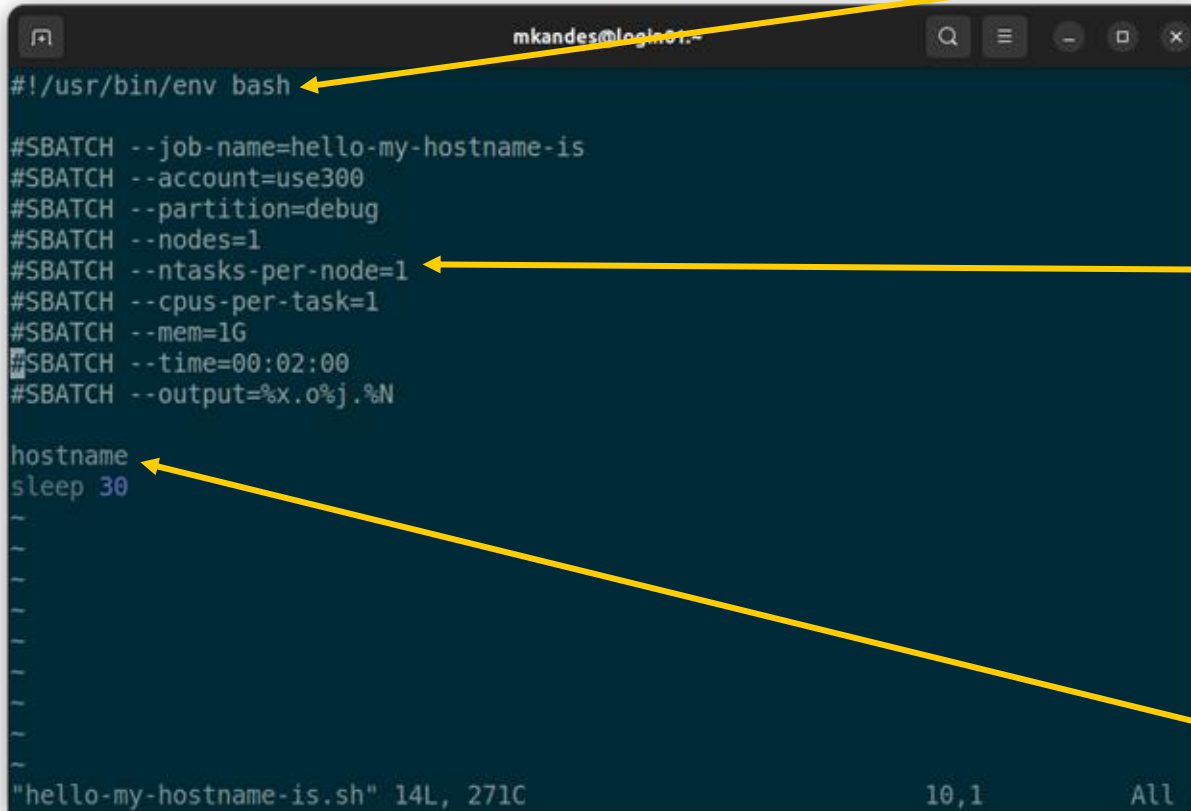
- Compile MPI hello world code:

```
[mthomas@login02 mpi]$  
[mthomas@login02 mpi]$ mpif90 -o hello_mpi  
hello_mpi.f90  
[mthomas@login02 mpi]$
```

```
[mthomas@login02 mpi]$  
[mthomas@login02 mpi]$ ls -lt hello_mpi  
-rwxr-xr-x 1 mthomas use300 22440 Jun 26  
18:45 hello_mpi  
[mthomas@login02 mpi]$
```

```
[mthomas@login02 mpi]$  
[mthomas@login02 mpi]$ sbatch hellompi-  
slurm.sb  
Submitted batch job 14124495  
[mthomas@login02 mpi]$
```

Exercise 1: Hello, my hostname is



```
#!/usr/bin/env bash

#SBATCH --job-name=hello-my-hostname-is
#SBATCH --account=use300
#SBATCH --partition=debug
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G
#SBATCH --time=00:02:00
#SBATCH --output=%x.o%j.%N

hostname
sleep 30
```

shebang (`#!`) character sequence is used to specify an interpreter that will execute the commands present in a plain text file.

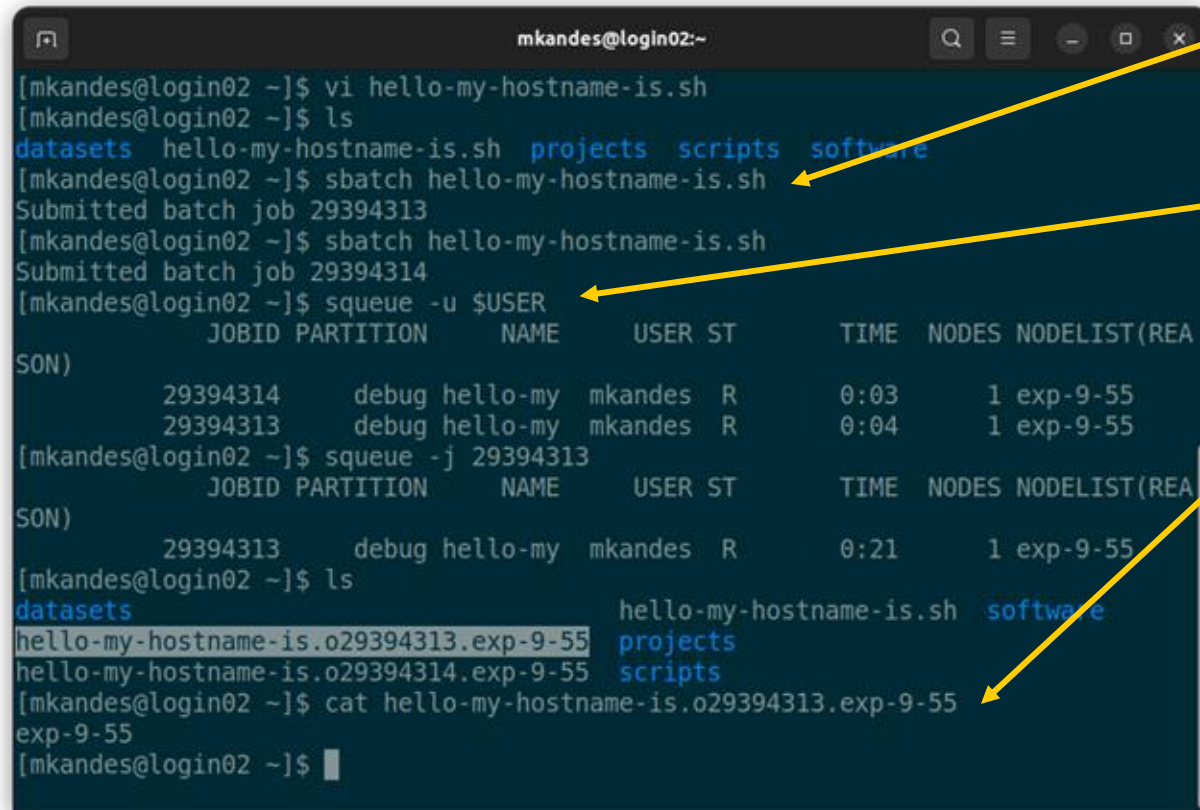
#SBATCH directives are used to specify the computational resource requirements for your job and various other attributes of it. e.g., name of the job and output files, which account to charge the job to, etc.

All executable commands to be run must follow the list of #SBATCH directives specified in any job script.

Exercise materials are available at <https://github.com/sdsc-complecs/batch-computing>

sbatch

The [sbatch](#) command is used to submit a batch job script to Slurm.



```
mkandes@login02:~  
[mkandes@login02 ~]$ vi hello-my-hostname-is.sh  
[mkandes@login02 ~]$ ls  
datasets hello-my-hostname-is.sh projects scripts software  
[mkandes@login02 ~]$ sbatch hello-my-hostname-is.sh  
Submitted batch job 29394313  
[mkandes@login02 ~]$ sbatch hello-my-hostname-is.sh  
Submitted batch job 29394314  
[mkandes@login02 ~]$ squeue -u $USER  
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)  
29394314 debug hello-my mkandes R 0:03 1 exp-9-55  
29394313 debug hello-my mkandes R 0:04 1 exp-9-55  
[mkandes@login02 ~]$ squeue -j 29394313  
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)  
29394313 debug hello-my mkandes R 0:21 1 exp-9-55  
[mkandes@login02 ~]$ ls  
datasets hello-my-hostname-is.sh software  
hello-my-hostname-is.o29394313.exp-9-55 projects  
hello-my-hostname-is.o29394314.exp-9-55 scripts  
[mkandes@login02 ~]$ cat hello-my-hostname-is.o29394313.exp-9-55  
exp-9-55  
[mkandes@login02 ~]$
```

Upon submission, every batch job is assigned a SLURM_JOB_ID number.

After submission, you can monitor the status of your jobs with the squeue command.

While a job runs, it will write standard output (and error) data to the file(s) named in your #SBATCH directives.

#SBATCH --output=%x.o%j.%N

%x = SLURM_JOB_NAME

%j = SLURM_JOB_ID

%N = hostname -s

Exercise 2: Roll 4 Pi

```
mkandes@login01:~  
#!/usr/bin/env bash  
#SBATCH --job-name=roll-4-pi  
#SBATCH --account=use300  
#SBATCH --partition=debug  
#SBATCH --nodes=1  
#SBATCH --ntasks-per-node=1  
#SBATCH --cpus-per-task=1  
#SBATCH --mem=1G  
#SBATCH --time=00:02:00  
#SBATCH --output=%x.o%j.%N  
  
module reset  
module load gcc/10.2.0  
module load python/3.8.12  
module list  
printenv  
  
time -p python3 pi.py 1000000000  
  
"roll-4-pi.sh" 19L, 359C 19,1 All
```

Exercise materials are available at <https://github.com/sdsc-complecs/batch-computing>

In most cases, you will want to follow the #SBATCH directives in your job script with a set of commands that configure the software environment in which the primary executable will run.

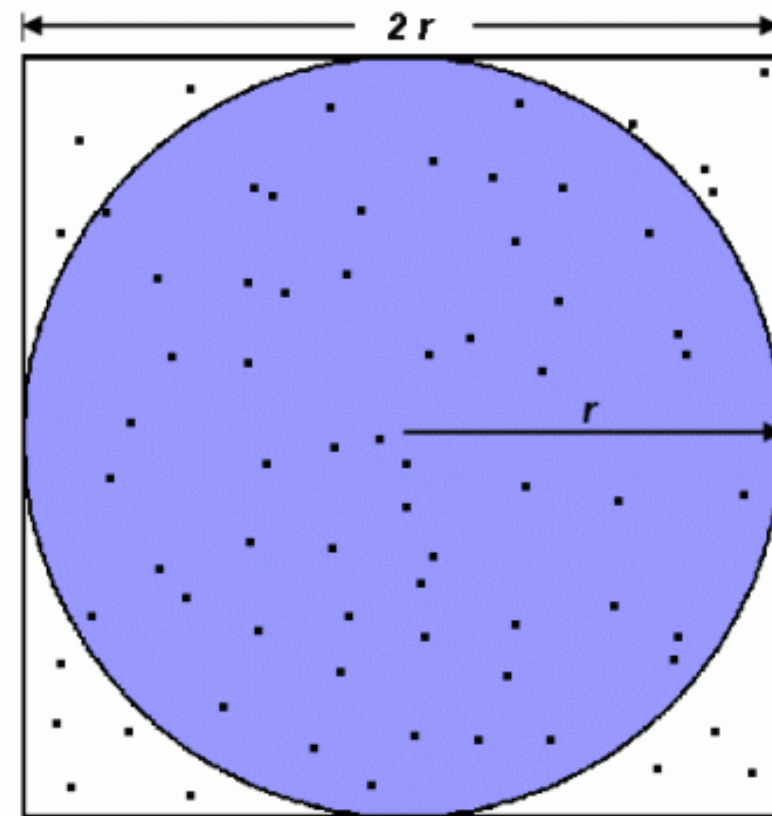
On many HPC systems, you will configure the software environment using pre-compiled and installed [software modules](#).

Here, we estimate the value of π using a [Monte Carlo method](#) implemented in Python.

The prepended [time](#) command measures the runtime of the estimation.

sbatch and roll (100M samples)

```
mkandes@login02:~$ ls
datasets          hello-my-hostname-is.sh  roll-4-pi.sh
hello-my-hostname-is.o29394313.exp-9-55  pi.py                   scripts
hello-my-hostname-is.o29394314.exp-9-55  projects                software
[mkandes@login02 ~]$ sinfo -p debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up       30:00    1    plnd exp-9-56
debug      up       30:00    1    mix  exp-9-55
[mkandes@login02 ~]$ sbatch roll-4-pi.sh
Submitted batch job 29402750
[mkandes@login02 ~]$ squeue -u $USER
squeue
JOBID PARTITION  NAME      USER ST       TIME  NODES NODELIST(REASON)
29402750  debug  roll-4-p  mkandes R          0:22    1 exp-9-55
[mkandes@login02 ~]$ squeue -u $USER
squeue
JOBID PARTITION  NAME      USER ST       TIME  NODES NODELIST(REASON)
29402750  debug  roll-4-p  mkandes R          0:22    1 exp-9-55
[mkandes@login02 ~]$ tail -n 4 roll-4-pi.o29402750.exp-9-55
3.1413662714136628
real 62.54
user 62.36
sys 0.01
[mkandes@login02 ~]$
```



$$A_S = (2r)^2 = 4r^2$$

$$A_C = \pi r^2$$

$$\pi = 4 \times \frac{A_C}{A_S}$$

What do you expect will happen if we increase the number of samples to one billion and resubmit the job??

sacct

The `sacct` command displays job data from the Slurm accounting database.

```
mkandes@login02:~  
hello-my-hostname-is.o29394314.exp-9-55 roll-4-pi.o29402750.exp-9-55  
hello-my-hostname-is.sh roll-4-pi.sh  
[mkandes@login02 ~]$ sinfo -p debug  
PARTITION AVAIL TIMELIMIT NODES STATE NODELIST  
debug      up      30:00      2  idle exp-9-[55-56]  
[mkandes@login02 ~]$ sbatch roll-4-pi.sh  
Submitted batch job 29408787  
[mkandes@login02 ~]$ squeue -u $USER  
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)  
29408787 debug roll-4-p mkandes R 1:19 1 exp-9-55  
[mkandes@login02 ~]$ tail -n 4 roll-4-pi.o29408787.exp-9-55  
BASH_FUNC_ml%=() { eval $(($LMOD_DIR/ml_cmd "$@" )  
}  
_=/usr/bin/printenv  
slurmstepd: error: *** JOB 29408787 ON exp-9-55 CANCELLED AT 2024-03-20T18:12:28  
DUE TO TIME LIMIT ***  
[mkandes@login02 ~]$ sacct -j 29408787  
JobID JobName Partition Account AllocCPUS State ExitCode  
-----  
29408787 roll-4-pi debug use300 1 TIMEOUT 0:0  
29408787.ba+ batch use300 1 CANCELLED 0:15  
29408787.ex+ extern use300 1 COMPLETED 0:0  
[mkandes@login02 ~]$
```

Use to look up information about completed and/or failed jobs when they are no longer visible in queue.

Quickly check the final state of a job and answer these questions: Did the job run and exit normally? If not, does Slurm know why it failed?

One of the most common failure modes for any batch job on an HPC system is running out of time.

Do **NOT** run extensive `sacct` queries without good reason.

What can you do when you submit a job you later realize has a mistake and/or might fail?



scancel

The [scancel](#) command is used to signal or cancel jobs, job arrays or job steps.

```
mkandes@login01:~  
hello-my-hostname-is.o29394314.exp-9-55  roll-4-pi.sh  
hello-my-hostname-is.sh                scripts  
pi.py                                   software  
projects  
[mkandes@login01 ~]$ sbatch roll-4-pi.sh  
Submitted batch job 29409705  
[mkandes@login01 ~]$ squeue -u $USER  
          JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REA  
SON)  
          29409705      debug  roll-4-p  mkandes  R         0:04        1 exp-9-55  
[mkandes@login01 ~]$ date  
Wed Mar 20 18:56:36 PDT 2024  
[mkandes@login01 ~]$ squeue -u $USER  
          JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REA  
SON)  
          29409705      debug  roll-4-p  mkandes  R         0:25        1 exp-9-55  
[mkandes@login01 ~]$ scancel 29409705  
[mkandes@login01 ~]$ tail -n 4 roll-4-pi.o29409705.exp-9-55  
BASH_FUNC_ml%%=( { eval $(($LMOD_DIR/ml_cmd "$@" )  
}  
_=/usr/bin/printenv  
slurmstepd: error: *** JOB 29409705 ON exp-9-55 CANCELLED AT 2024-03-20T18:56:47  
***  
[mkandes@login01 ~]$
```

Use whenever you want to remove a pending job from the squeue or stop a running job.

Table of Contents: Part 1

- High-Performance Computing Hardware and System Architecture
- Batch Job Scheduling and Resource Management
- Getting Started with the Slurm Workload Manager
 - `squeue` - view information about jobs located in the Slurm scheduling queue
 - `sinfo` - view information about Slurm nodes and partitions
- First Batch Job(s) with Slurm
 - `sbatch` - submit a batch script to Slurm
 - `scancel` - used to signal or cancel jobs, job arrays or job steps
 - `sacct` - displays accounting data for all jobs in the Slurm database
- **Best Practices with Slurm**

Parallel Computing Myth #2: Scheduler Variant

Simply requesting more compute resources from your batch job scheduler will automatically reduce the time to solution for your problem.

WRONG!

The truth is your application must be designed to take advantage of the additional compute resources you've requested. If it is designed to do so, then you generally must still configure your batch job to correctly utilize those resources.

Prerequisite Knowledge:

- [Parallel Computing Concepts](#)



Exercise 3: Build Job, Run Job

If you need to compile and install software on an HPC system, then you should consider writing a batch job to build and install the software instead of proceeding to do so interactively from the command-line.

There are a number of advantages of using this approach:

- build job script will document how to build and install the software
- build and install logs will be available in the job's standard output (and/or error) files
- software may need to be compiled on a specific compute node architecture
- software environment created within the build job to compile the software will need to be replicated in the job scripts that eventually run the software anyway

Exercise materials are available at <https://github.com/sdsc-complecs/batch-computing>

Parallel Pi with OpenMP

Parallel computing jobs can take advantage of more CPU resources

[illegible]

Build Job

```
mkandes@login01:~  
#!/usr/bin/env bash  
#SBATCH --job-name=run-pi-omp  
#SBATCH --account=use300  
#SBATCH --partition=debug  
#SBATCH --nodes=1  
#SBATCH --ntasks-per-node=1  
#SBATCH --cpus-per-task=4  
#SBATCH --mem=1G  
#SBATCH --time=00:02:00  
#SBATCH --output=%x.o%j.%N  
  
module reset  
module load gcc/10.2.0  
module list  
export OMP_NUM_THREADS="${SLURM_CPUS_PER_TASK}"  
printenv  
  
time -p ./pi_omp.x -s 10000000000  
~  
~  
~  
~  
"run-pi-omp.sh" 19L, 383C
```

Run Job

Use Slurm environment variables!

Exercise 4: GPU-Accelerated Workloads

```
mkandes@login02:~  
#!/usr/bin/env bash  
  
#SBATCH --job-name=train-cnn-cifar-c10-fp32-e42-bs256-tensorflow-22.08-tf2-py3-1v100  
#SBATCH --account=use300  
#SBATCH --partition=gpu-debug  
#SBATCH --nodes=1  
#SBATCH --ntasks-per-node=10  
#SBATCH --cpus-per-task=1  
#SBATCH --mem=92G  
#SBATCH --gpus=1  
#SBATCH --time=00:05:00  
#SBATCH --output=%x.o%j.%N  
  
declare -xr LOCAL_SCRATCH_DIR="/scratch/${USER}/job_${SLURM_JOB_ID}"  
declare -xr LUSTRE_PROJECTS_DIR="/exppanse/lustre/projects/${SLURM_JOB_ACCOUNT}/${USER}"  
declare -xr LUSTRE_SCRATCH_DIR="/exppanse/lustre/scratch/${USER}/temp_project"  
declare -xr SINGULARITY_CONTAINER_DIR="/cm/shared/apps/containers/singularity"  
  
declare -xr SINGULARITY_MODULE='singularitypro/3.9'  
  
module purge  
module load "${SINGULARITY_MODULE}"  
module list  
export KERAS_HOME="${LOCAL_SCRATCH_DIR}"  
printenv  
  
time -p singularity exec --bind "${KERAS_HOME}:/tmp" --nv "${SINGULARITY_CONTAINER_DIR}/tensorflow/tensorflow_22.08-tf2-py3.sif" \  
python3 -u tf2-train-cnn-cifar.py --classes 10 --precision fp32 --epochs 42 --batch_size 256
```

GPUs are now a first-class compute resource in Slurm

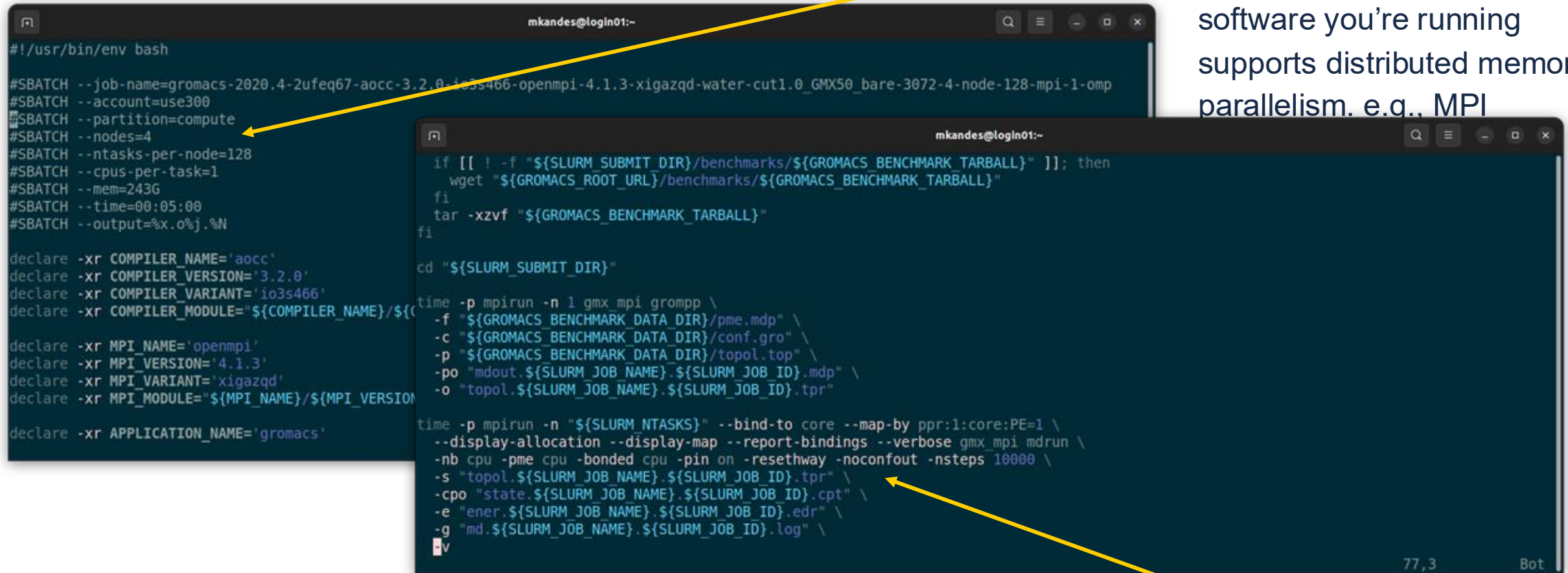
Don't forget to define your own environment variables as needed!

Exercise materials are available at <https://github.com/sdsc-complecs/batch-computing>

Your job's software environment can be a mix of standard HPC software modules and alternative software package management methods. e.g., Singularity containers, conda environments

Exercise 5: Multinode MPI Jobs

You can request multiple compute nodes for your jobs. But only do so if the software you're running supports distributed memory parallelism. e.g., MPI



```
#!/usr/bin/env bash

#SBATCH --job-name=gromacs-2020.4-2ufeq67-aocc-3.2.0-103s466-openmpi-4.1.3-xigazqd-water-cut1.0_GMX50_bare-3072-4-node-128-mpi-1-omp
#SBATCH --account=use300
#SBATCH --partition=compute
#SBATCH --nodes=4
#SBATCH --ntasks-per-node=128
#SBATCH --cpus-per-task=1
#SBATCH --mem=243G
#SBATCH --time=00:05:00
#SBATCH --output=%x.o%j.%N

declare -xr COMPILER_NAME='aocc'
declare -xr COMPILER_VERSION='3.2.0'
declare -xr COMPILER_VARIANT='io3s466'
declare -xr COMPILER_MODULE="${COMPILER_NAME}/${COMPILER_VERSION}-${COMPILER_VARIANT}"

declare -xr MPI_NAME='openmpi'
declare -xr MPI_VERSION='4.1.3'
declare -xr MPI_VARIANT='xigazqd'
declare -xr MPI_MODULE="${MPI_NAME}/${MPI_VERSION}-${MPI_VARIANT}"

declare -xr APPLICATION_NAME='gromacs'

if [[ ! -f "${SLURM_SUBMIT_DIR}/benchmarks/${GROMACS_BENCHMARK_TARBALL}" ]]; then
    wget "${GROMACS_ROOT_URL}/benchmarks/${GROMACS_BENCHMARK_TARBALL}"
fi
tar -xzf "${GROMACS_BENCHMARK_TARBALL}"
fi

cd "${SLURM_SUBMIT_DIR}"

time -p mpirun -n 1 gmx_mpi grompp \
-f "${GROMACS_BENCHMARK_DATA_DIR}/pme.mdp" \
-c "${GROMACS_BENCHMARK_DATA_DIR}/conf.gro" \
-p "${GROMACS_BENCHMARK_DATA_DIR}/topol.top" \
-po "mdout.${SLURM_JOB_NAME}.${SLURM_JOB_ID}.mdp" \
-o "topol.${SLURM_JOB_NAME}.${SLURM_JOB_ID}.tpr"

time -p mpirun -n "${SLURM_NTASKS}" --bind-to core --map-by ppr:1:core:PE=1 \
--display-allocation --display-map --report-bindings --verbose gmx_mpi mdrun \
-nb cpu -pme cpu -bonded cpu -pin on -resetway -noconfout -nsteps 10000 \
-s "topol.${SLURM_JOB_NAME}.${SLURM_JOB_ID}.tpr" \
-cpo "state.${SLURM_JOB_NAME}.${SLURM_JOB_ID}.cpt" \
-e "ener.${SLURM_JOB_NAME}.${SLURM_JOB_ID}.edr" \
-g "md.${SLURM_JOB_NAME}.${SLURM_JOB_ID}.log" \
-v
```

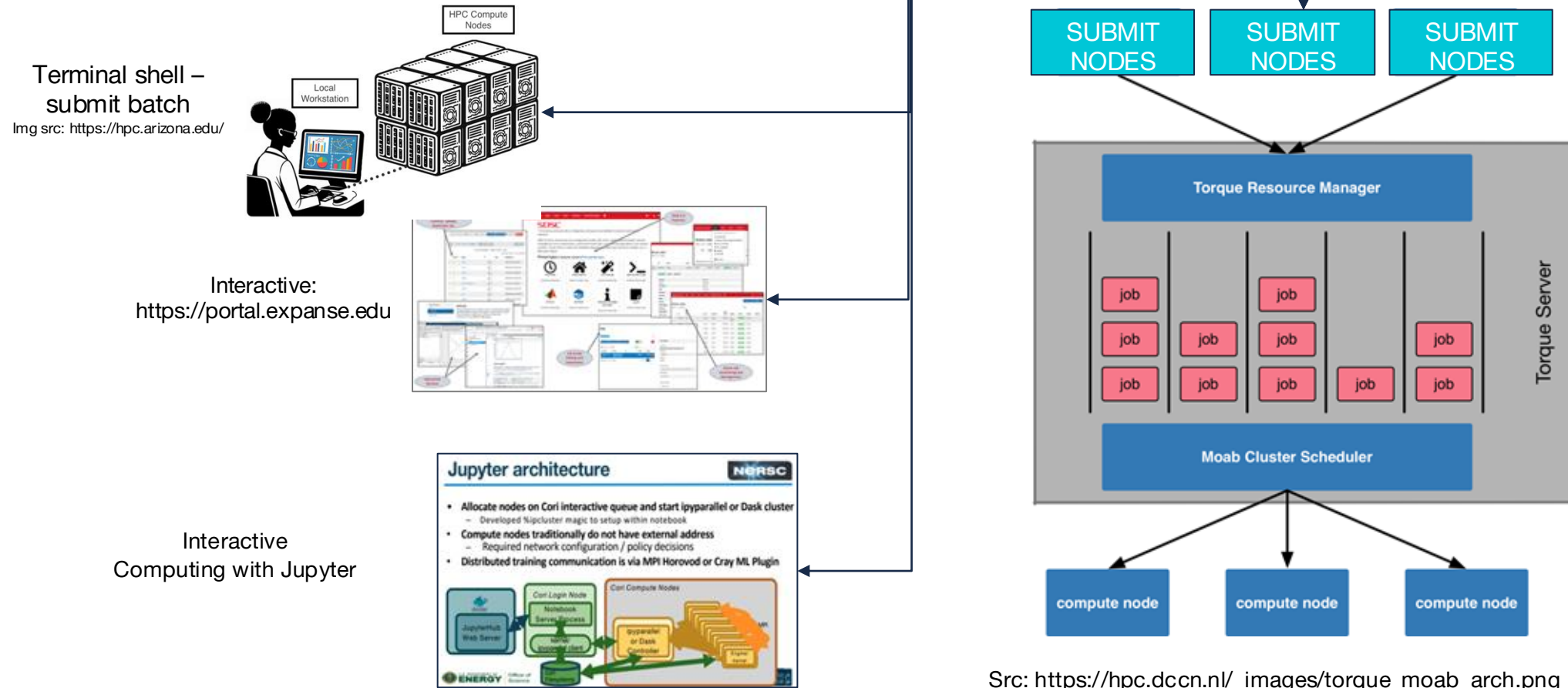
Exercise materials are available at <https://github.com/sdsc-complecs/batch-computing>

But again, you will still likely need to configure the call to the application to use the requested resources correctly.

Table of Contents: Part 2

- **Defining Interactive HPC Computing**
- **Accessing Interactive HPC Nodes**
- **Examples of Interactive Applications**
- **Secure Jupyter Notebooks**
- **Expanse Portal**
- **Questions & Answers**

HPC Jobs can be run as batch jobs (background) or interactively (real time)



What is Interactive HPC Computing

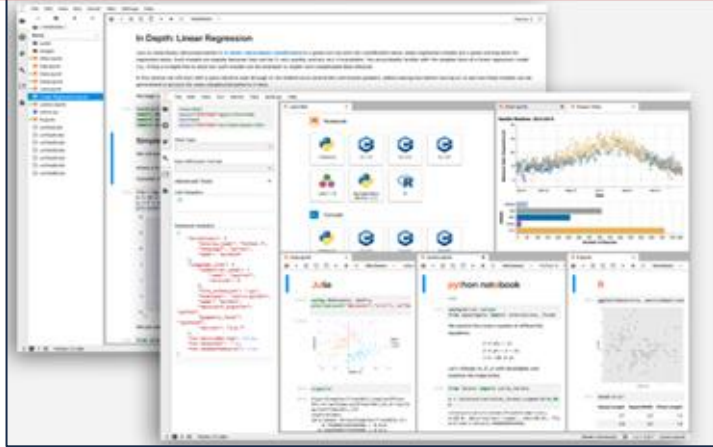
- In **computer** science, **interactive computing** refers to software which accepts input from the user as it runs.
 - **Interactive** software includes commonly used programs, such as word processors or spreadsheet applications.
- **Interactive HPC computing** involves *real-time* user inputs to perform tasks on a set of compute node(s) including:
 - Code development, real-time data exploration, and visualizations.
 - Used when applications have large data sets or are too large to download to local device, software is difficult install, etc.
 - User inputs come via command line interface or application GUI (Jupyter Notebooks, Matlab, R-studio).
 - Actions performed on remote compute nodes as a result of user input or program output.

Interactive HPC Computing- Motivation

- **Need more memory:** Your jobs no longer fit onto the CPU/system you have been using:
 - That is why Expanse has 768 nodes (128 cores per node), with 256 GB DDR memory on each node
 - My MacBook Pro: 1 node, 8 cores, 16 GB DDR
- **Too much data:** your application needs more room:
 - Expanse has 1TB NVME/node, and 12 PB file system.
 - My MBP: 500 GB, no NVME
- **Your network is too slow:**
 - Expanse has connections to ~ 150GB/sec (or faster) networks
 - My MBP: 300 Mbps download; 11 Mbps upload
 - Bioinfo lab, running analysis on PC: FastQ dataset ~ 500 MB: would need 360+ seconds to upload 1 run.

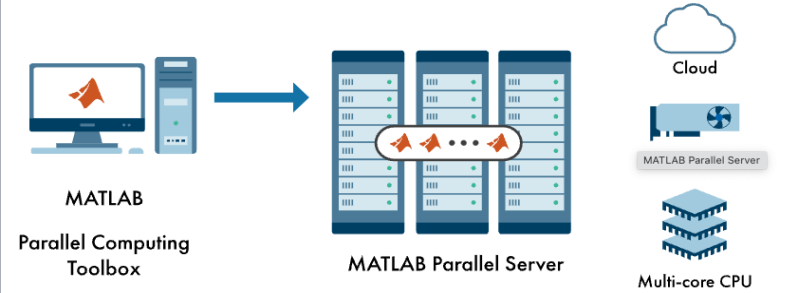
Interactive HPC Computing

Jupyter Notebook: <https://jupyter.org/>



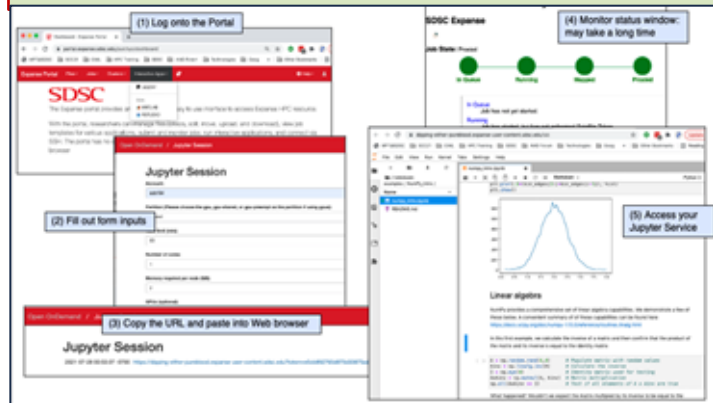
Parallel Matlab (Mathworks)

```
parpool("HPC",1000);  
  
parfor ii = 1:3000  
    c(ii,:) = eig(rand(1000));  
end
```



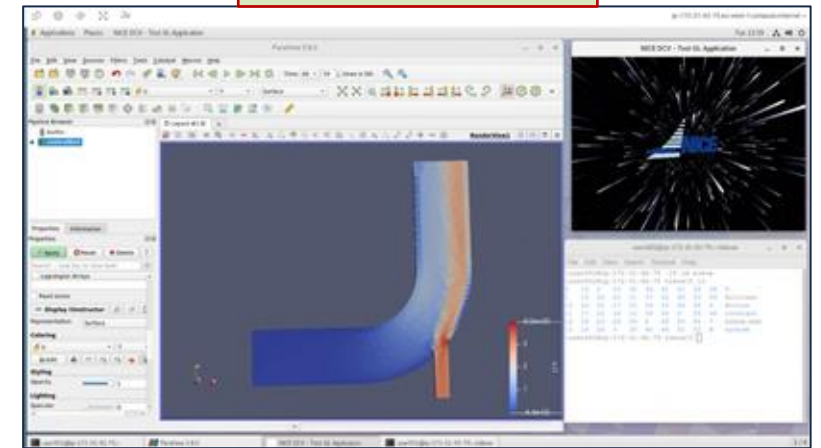
<https://azuremarketplace.microsoft.com/en-us/marketplace/apps/mathworks-inc.matlab-parallel-server-listing?tab=Overview>

Expansive User Portal



<https://portal.expansive.sdsc.edu>

Paraview (AWS)



<https://aws.amazon.com/blogs/compute/how-to-run-3d-interactive-applications-with-nice-dcv-in-aws-batch/>

Examples of Interactive HPC Methods & Applications

- Applications/Data Analysis Platforms:
 - Support interactions from within the code/model using libraries
 - Matlab, R, Jupyter Notebooks
 - NetCDF, HPF, TAU, ParaView
 - other
- Interact with large data sets after job is done:
 - Unix: query file info, location, output, grep, awk, sed
 - Cat the file contents from batch job or raw data
 - NetCDF data browser

Note: Interactive Services have a Key Vulnerability: They Provide Access to HPC File Systems, often over HTTP

SDSC Interactive Services Policy:

- Portals, JupyterHub, and other services cannot be mounted directly to disk (must be on VM or external):
 - Many use root in vulnerable ways
 - If a user launches Jupyter Lab or Notebooks, the jobs will be killed.
- Applications cannot run on login nodes.

SDSC recommendation:

- Use secure connections: when you choose insecure connections your account is vulnerable to hacking.
- Use portal.expense.sdsc.edu

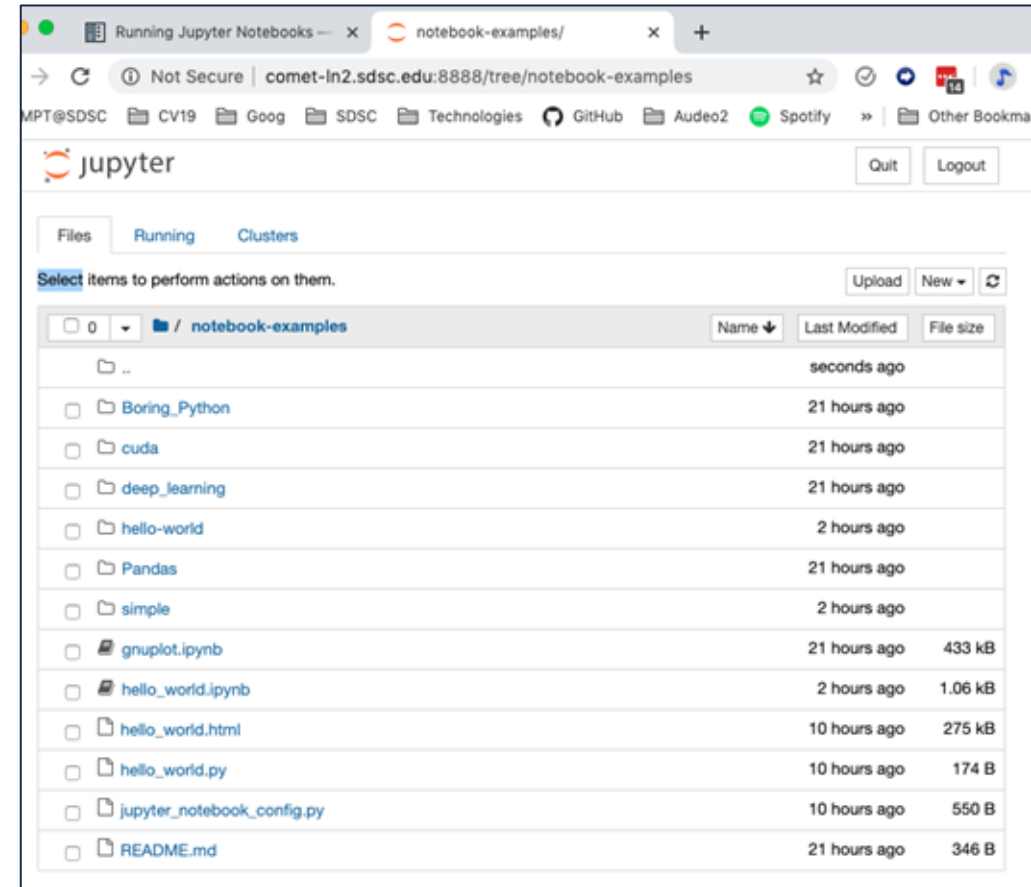


Table of Contents: Part 2

- Defining Interactive HPC Computing
- **Accessing Interactive HPC Nodes**
- Examples of Interactive Applications
- Secure Jupyter Notebooks
- Expanse Portal
- Questions & Answers

When would you need to use an Interactive node?

- The application is interactive (Matlab, Notebooks)
- Need to stage large amounts of data
- AI applications (Pytorch, Tensorflow)
- Need to compile a very large, complex application that may take a long time.
 - Java code, 1 million lines, > 30 minutes on DELL 5511 laptop

Accessing Interactive Compute Nodes on Expanse

- Connect to HPC system (e.g. Expanse) via terminal using *SSH* secure connections
- Use the *srun* command to obtain nodes for ‘live,’ command line interactive access:

CPU	<code>srun --partition=debug --pty --account=abc123 --nodes=1 --ntasks-per-node=4 --mem=8G -t 00:30:00 --wait=0 --export=ALL /bin/bash</code>
GPU	<code>srun --partition=gpu-debug --pty --account=use300 --ntasks-per-node=10 --nodes=1 --mem=96G --gpus=1 -t 00:30:00 --wait=0 --export=ALL /bin/bash</code>

(Tested 04/17/2024)

Using An Interactive CPU node

```
[mthomas@login01 calc-prime]$ srun --partition=debug --pty --account=<<project>> -  
-nodes=1 --ntasks-per-node=4 --mem=8G -t 00:30:00 --wait=0 --export=ALL /bin/bash
```

```
srun: job 24457429 has been allocated resources
```

```
[mthomas@exp-9-55 calc-prime]$ module purge
```

```
[mthomas@exp-9-55 calc-prime]$ module load slurm
```

```
[mthomas@exp-9-55 calc-prime]$ module load cpu
```

```
[mthomas@exp-9-55 calc-prime]$ module load gcc/10.2.0
```

```
[mthomas@exp-9-55 calc-prime]$ module load openmpi/4.1.1
```

```
[mthomas@exp-9-55 calc-prime]$ mpirun -n 16 ./mpi_prime
```

```
06 August 2023 11:10:26 PM
```

```
PRIME_MPI n_hi= 5000000 C/MPI version
```

```
An MPI example program to count the number of primes: # processes is 64
```

N	Pi	Time
1	0	0.013258
2	1	0.001058
4	2	0.000101
8	4	0.000101

```
[SNIP]
```

131072	12251	0.110848
262144	23000	0.410792
524288	43390	1.527210
1048576	82025	5.733612
2097152	155611	21.725862

```
PRIME_MPI - Master process: Normal end of execution.
```

```
06 August 2023 11:12:26 PM
```

Request an interactive node
for 30 minutes

- Exit interactive session when your work is done or you will be charged more CPU time.
- Beware of oversubscribing your job: don't ask for more cores than you have requested.
- Intel compiler allows this, but your performance will be degraded.

https://www.sdsc.edu/systems/expanse/user_guide.html

Using Interactive GPU nodes

```
[snip]
Last login: Fri Feb 18 12:58:32 2022 from 76.176.117.51
[username@login02 ~]$
[username@login02 ~]$ srun --partition=gpu-debug --pty --account=use300 --ntasks-per-node=10 --nodes=1 --
mem=96G --gpus=1 -t 00:30:00 --wait=0 --export=ALL /bin/bash
srun: job 9794018 queued and waiting for resources
srun: job 9794018 has been allocated resources
[mthomas@exp-14-57 ~]$
[mthomas@exp-14-57 ~]$ nvidia-smi
Fri Feb 18 13:04:19 2022

+-----+
| NVIDIA-SMI 460.32.03    Driver Version: 460.32.03    CUDA Version: 11.2    |
+-----+-----+-----+-----+
| GPU Name      Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|      Memory-Usage | GPU-Util  Compute M. |
|               |              MIG M. |                    |
+=====+
| 0 Tesla V100-SXM2...  On   | 00000000:86:00:0 Off |          0          |
| N/A   34C   P0   41W / 300W |  0MiB / 32510MiB |    0%    Default   |
|               |              N/A   |                    |
+-----+-----+-----+-----+

+-----+
| Processes:                                                       |
| GPU  GI  CI       PID Type Process name                      GPU Memory |
|   ID ID                  Usage   |
+=====+
| No running processes found                                         |
+-----+
[username@exp-14-57 ~]$ exit
```

Request an interactive node
for 30 minutes

Verify you are on a GPU node

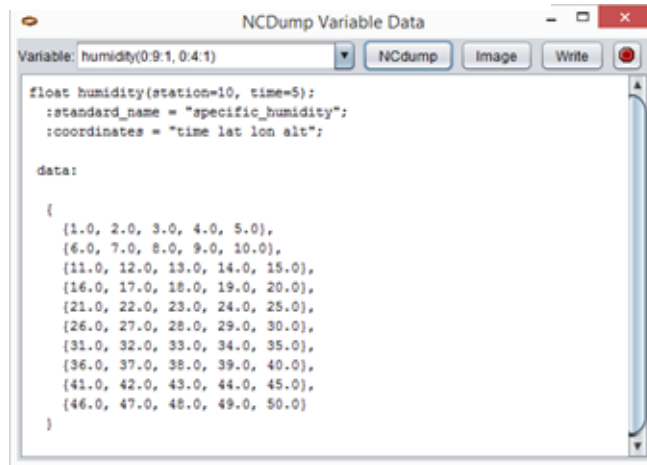
Exit when tasks are done

Table of Contents: Part 2

- Defining Interactive HPC Computing
- Accessing Interactive HPC Nodes
- **Examples of Interactive Applications**
- Secure Jupyter Notebooks
- Expanse Portal
- Questions & Answers

NetCDF Data Files

- NetCDF (Network Common Data Form) files are binary files containing both data and metadata about the data.
- API: software libraries and machine-independent data formats that support the creation, access, and sharing of array-oriented scientific data
- Similar to high-density format (HDF)
- NetCDF clients (*ncview*, *ncdump*) can be used to query and plot data in real-time



NCDump Variable Data

Variable: humidity(0.9:1, 0.4:1)

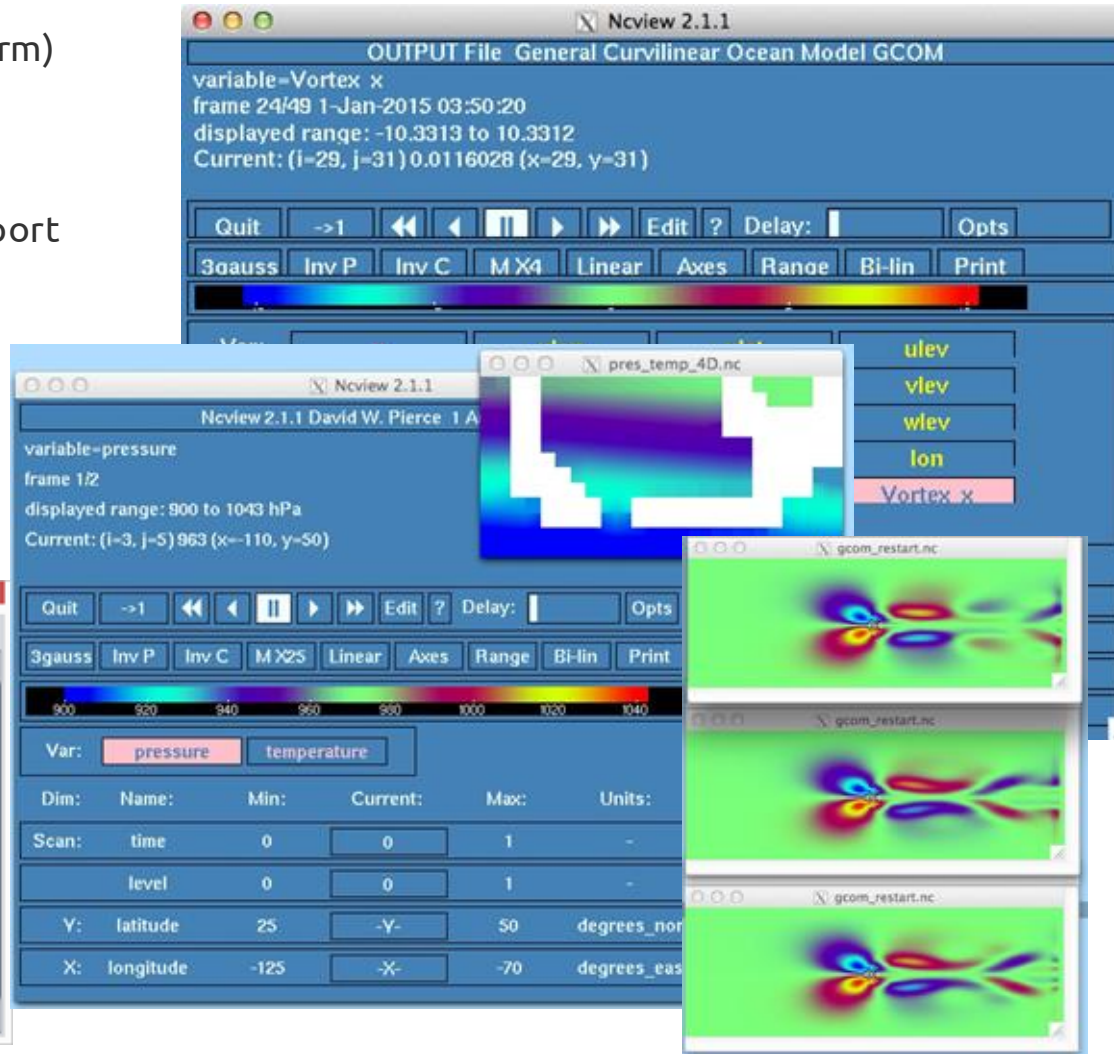
NCdump Image Write

```
float humidity(station=10, time=5);
:standard_name = "specific_humidity";
:coordinates = "time lat lon alt";

data:

[
  {1.0, 2.0, 3.0, 4.0, 5.0},
  {6.0, 7.0, 8.0, 9.0, 10.0},
  {11.0, 12.0, 13.0, 14.0, 15.0},
  {16.0, 17.0, 18.0, 19.0, 20.0},
  {21.0, 22.0, 23.0, 24.0, 25.0},
  {26.0, 27.0, 28.0, 29.0, 30.0},
  {31.0, 32.0, 33.0, 34.0, 35.0},
  {36.0, 37.0, 38.0, 39.0, 40.0},
  {41.0, 42.0, 43.0, 44.0, 45.0},
  {46.0, 47.0, 48.0, 49.0, 50.0}
]
```

<https://unidata.ucar.edu/netcdf>



Outline

- What is High-Performance Computing (HPC)?
- What is HPC batch computing
- Defining Interactive HPC Computing
- Accessing Interactive HPC Nodes
- **Interactive Application Examples**
 - Viewing Data: unix file operators (grep, awk, cat), gnuplot, NetCDF
 - **Programming & Visualization Platforms: Matlab, R, Jupyter Notebooks (and NetCDF)**
 - Gateways & Portals: simplify access to interactive apps

Expanse Interactive Jobs: Running Matlab from Command Line

```
[mthomas@login01 ~]$ srun --partition=gpu-debug --pty --account=use300 --ntasks-per-node=10 --  
nodes=1 --mem=96G --gpus=1 -t 00:30:00 --wait=0 --export=ALL /bin/bash
```

```
srun: job 14833549 queued and waiting for resources
```

```
srun: job 14833549 has been allocated resources
```

```
[mthomas@exp-9-55 ~]$ module load gpu/0.17.3b
```

```
[mthomas@exp-9-55 ~]$ module load matlab/2022b/nmbr5dd
```

```
[mthomas@exp-9-55 ~]$ module list
```

```
Currently Loaded Modules:
```

```
1) shared          3) sdsc/1.0      5) gpu/0.17.3b    (g)  
2) slurm/expanse/21.08.8 4) DefaultModules 6) matlab/2022b/nmbr5dd
```

Check that you have the right modules loaded

```
[mthomas@exp-9-55 ~]$ matlab
```

```
MATLAB is selecting SOFTWARE OPENGL  
rendering.
```

```
< M A T L A B (R) >
```

```
Copyright 1984-2022 The MathWorks, Inc.
```

```
R2022a (9.12.0.1884302) 64-bit (glnxa64)
```

```
February 16, 2022
```

```
>> a = [1 3 5; 2 4 6; 7 8 10]
```

```
a =
```

```
1   3   5  
2   4   6  
7   8  10
```

```
>> b=sin(a)
```

```
b =
```

```
0.8415  0.1411 -0.9589  
0.9093 -0.7568 -0.2794  
0.6570  0.9894 -0.5440
```

```
>> plot(b)  -- nothing happens
```

```
>>
```


Expanse Interactive Jobs: Running Matlab with GUI

To use a Graphical User Interface (GUI) as part of your interactive job, **you will need to set up Xforwarding**. Example below is for using XQuartz on a MAC. For examples for Windows, see: <http://systems.eecs.tufts.edu/x11-forwarding/>

Step 1: Log on to Expanse, using **-Y** option (trusted); create two connections; **W2 should be the XQuartz terminal**

```
[mthomas@home]$ ssh -Y mthomas@login.expanse.sdsc.edu
```

W1

```
[mthomas@home]$ ssh -Y mthomas@login.expanse.sdsc.edu
```

W2

Step 2: Window 1: Request an interactive node, using **srun command**
This window will not be used for anything else but holding the interactive node connection

```
[mthomas@login01 ~]$ srun --partition=debug --pty --account=use300 --nodes=1 --ntasks-per-node=4 --mem=8G -t 00:30:00 --wait=0 --export=ALL /bin/bash  
srun: job 26814162 queued and waiting for resources  
srun: job 26814162 has been allocated resources  
[mthomas@exp-9-55 ~]$
```

W1

Step 3: Window 2:
Connect to the interactive node: **atlab**

```
[mthomas@login02 ~]$ ssh -Y exp-9-55  
Last login: Wed Dec 6 19:23:47 2023 from 10.21.0.19
```

W2

Step 4: Window 2:
Setup Matlab module environment
Launch Matlab

```
[mthomas@exp-9-55 ~]$ module load gpu/0.17.3b  
[mthomas@exp-9-55 ~]$ module load matlab/2022b/nmbr5dd  
[mthomas@exp-9-55 ~]$ module list  
Currently Loaded Modules:  
1) shared 3) sdsc/1.0 5) gpu/0.17.3b (g)  
2) slurm/expanse/21.08.8 4) DefaultModules 6) matlab/2022b/nmbr5dd  
[mthomas@exp-9-55 ~]$ matlab  
MATLAB is selecting SOFTWARE OpenGL rendering.
```

W2

GUI :
USE EXPANSE PORTAL

HPC Interactive Jobs: Running Matlab with GUI

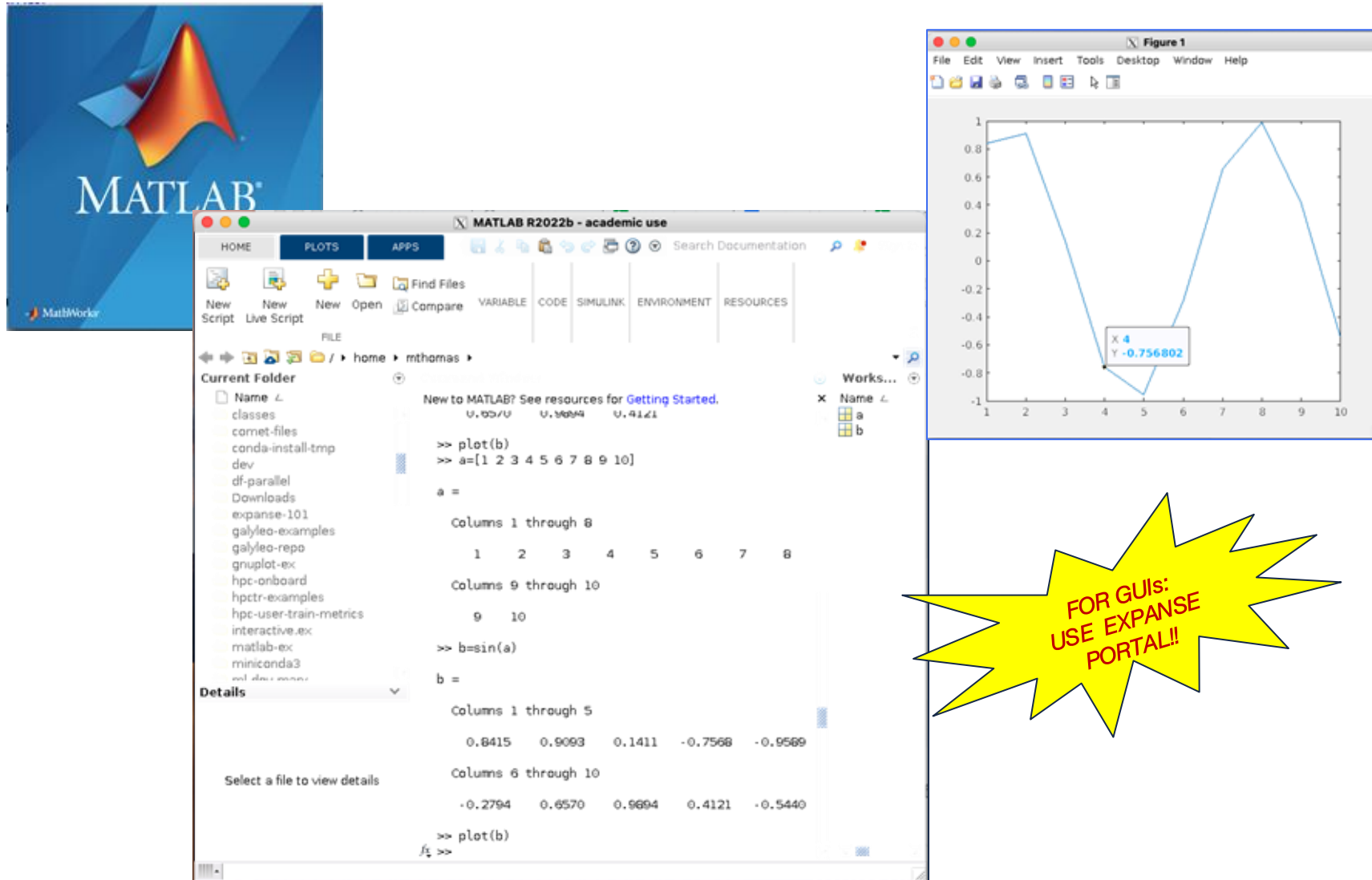


Table of Contents: Part 2

- Defining Interactive HPC Computing
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- **Secure Jupyter Notebooks**
- Expanse Portal
- Questions & Answers

Jupyter Notebooks

What is Jupyter?

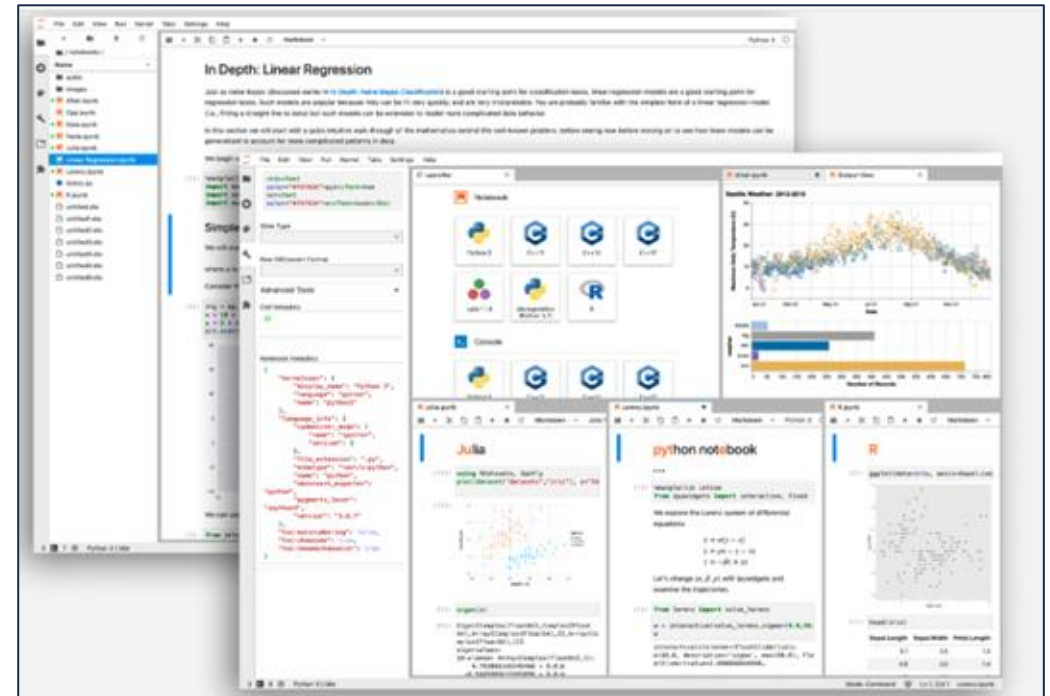
*Jupyter is a free, open-source, **interactive** web tool known as a computational notebook, which researchers can use to combine software code, computational output, explanatory text and multimedia resources in a single document. (J. Perkel, <https://www.nature.com/articles/d41586-018-07196-1>)*

Common Jupyter Services:

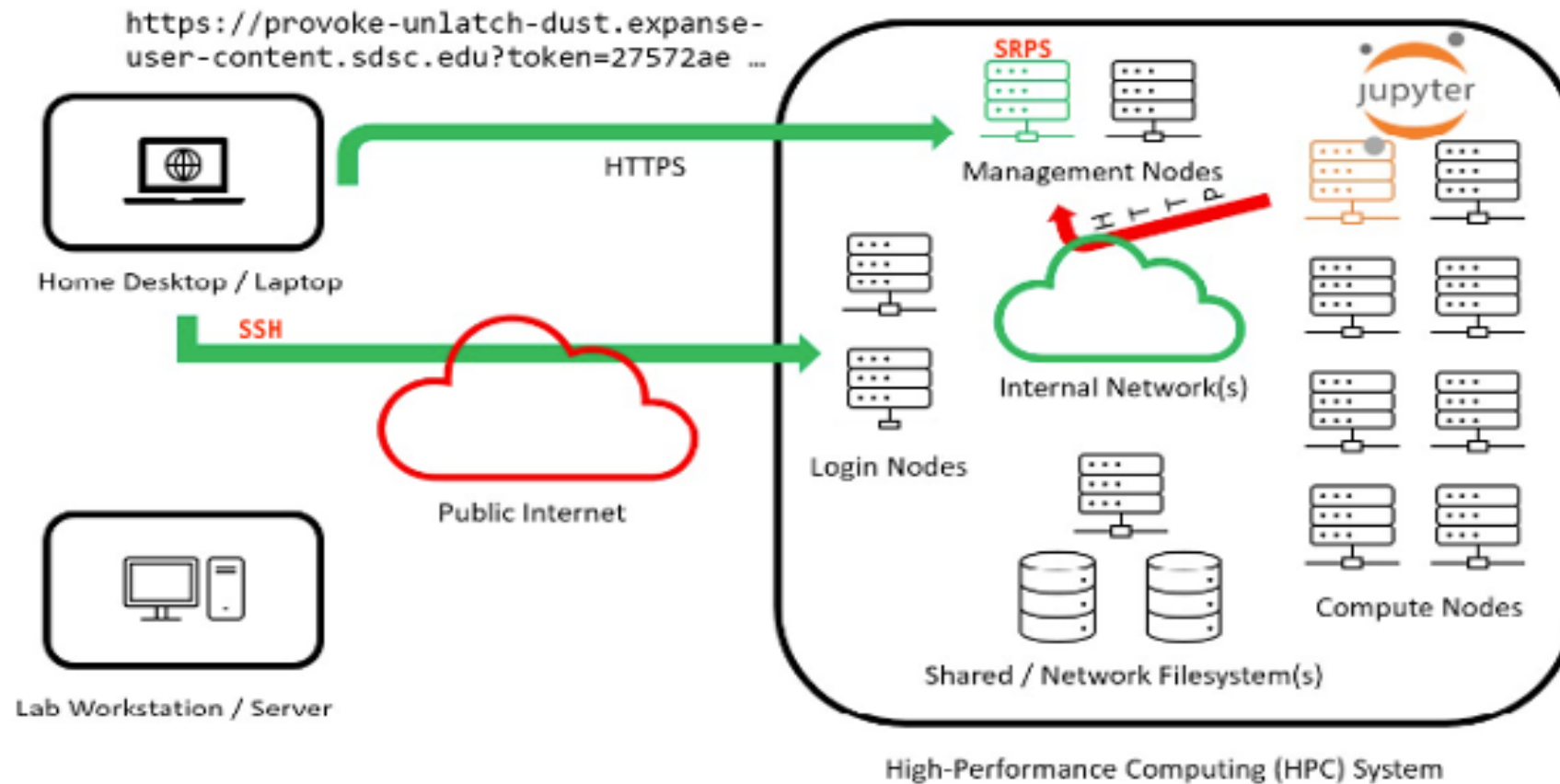
- Jupyter Notebooks (single user)
- JupyterLab: advanced version of notebook
- JupyterHub: multiuser Jupyter service

Running **Secure** Notebooks HPC Systems

- Install notebook application:
 - Locally: install Anaconda on your laptop (default is **HTTP**)
- Connecting local host (e.g. laptop) to HPC:
 - HTTP (default, insecure)
 - HTTP + SSH tunneling (secure, but inconvenient)
 - **HTTPS + using the *Satellite Reverse Proxy Service (SRPS)* and *galileo* client (secure, convenient)**
- You can launch Jupyter services on SDSC:
 - CPU and GPUs
 - Interactive nodes: command line or Slurm batch script
- **Treat the Notebook URL like a Password!**



Running Notebooks Remotely & Securely Using SRPS & **galileo**



Src: M Kandes: https://education.sdsc.edu/training/interactive/202112_running_jupyter_notebooks_on_expance/

galileo

- 2nd generation shell utility developed to orchestrate a user's interaction with both Satellite and Slurm to start a Jupyter session within a batch job.
- Developed while reviewing start-jupyter (prototype client) codebase to sort out how best to support Expanse (OOD) Portal and HPC User Services Group long-term; integrated into an existing SSH tunneling orchestration utility to use Satellite proxy service instead
- **Key features** in design:
 - HTTPS URL
 - Supports containers (Singularity on Expanse)
 - No need to install conda environment or update packages
 - Increases flexibility for users to configure software environment; but also try to makes it simpler for them to do this themselves
 - Batch job script is generated completely on-the-fly.
 - Command-line argument driven.
 - Quiet mode for OOD portal

<https://github.com/mkandes/galileo>

Satellite Client: galileo

Key features in design:

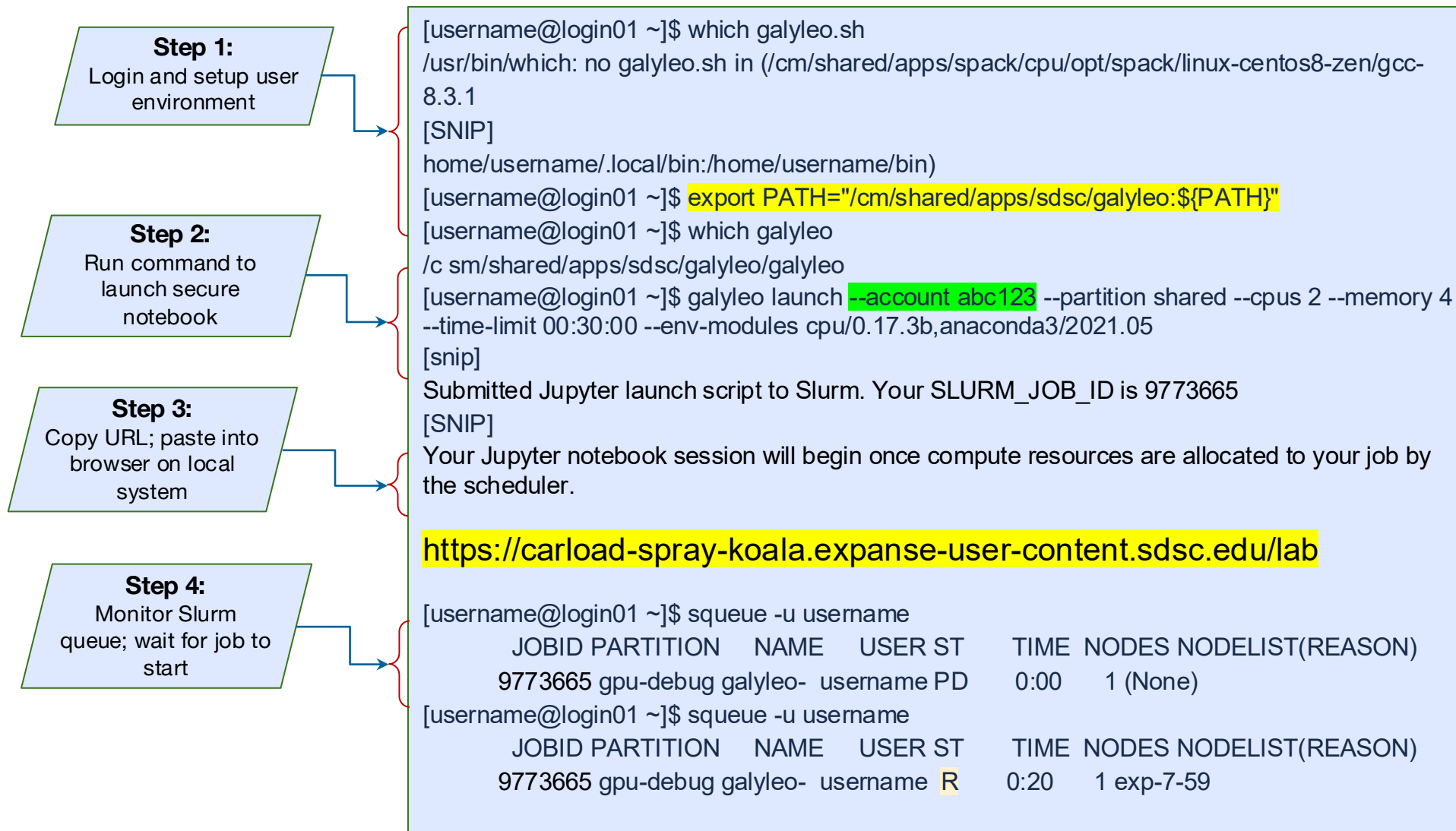
- User calls galileo.sh launch script, which requests token from Satellite, passes token to batch job script and submits the job to Slurm; token redeemed from batch job once it runs
- Increase flexibility for users to configure software environment; but also try to make it simpler for them to do themselves
- Batch job script is generated completely on-the-fly.
- Command-line argument driven.
- Quiet mode for OOD portal

```
[username@login01 ~]export PATH="/cm/shared/apps/sdsc/galileo:${PATH}"

[username@login01 ~]$ galileo.sh --help
USAGE: galileo.sh launch [command-line option] {value}
command-line option    : value
-A | --account         :
-R | --reservation     :
-p | --partition       :
-q | --qos              :
-N | --nodes           :
-n | --ntasks-per-node :
-c | --cpus-per-task   :
-M | --memory-per-node : GB
-m | --memory-per-cpu  : GB
-G | --gpus            :
  | --gres             :
-t | --time-limit      :
-j | --jupyter         :
-d | --notebook-dir    :
-r | --reverse-proxy   :
-D | --dns-domain      :
-s | --sif             :
-B | --bind            :
  | --nv               :
-e | --env-modules     :
  | --conda-env        :
-Q | --quiet           : 1
```

<https://github.com/sdsc/galileo>

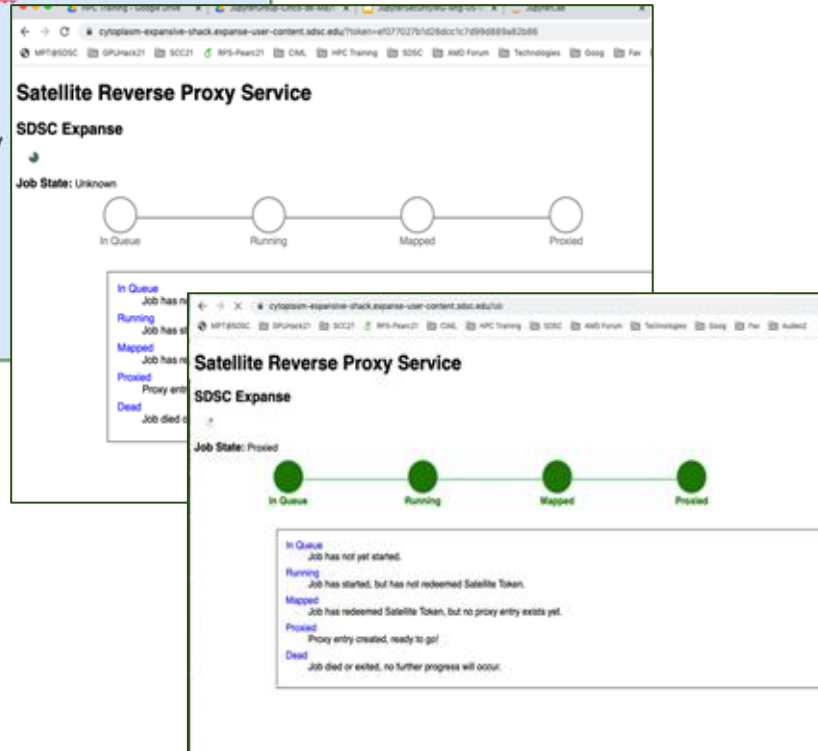
Launching CPU notebooks using galileo



Satellite-Galileo System

```
[username@login01 ~]export
PATH="/cm/shared/apps/sdsc/galileo:${PATH}"

[username@login01 ~]$ galileo.sh --help
USAGE: galileo.sh launch [command-line
option] {value}
command-line option      : value
-A | --account           :
-R | --reservation       :
-p | --partition         :
-q | --qos                :
-N | --nodes             :
-n | --ntasks-per-node   :
-c | --cpus-per-task     :
-M | --memory-per-node   : GB
-m | --memory-per-cpu    : GB
-G | --qos               :
-g | --gres              :
-t | --time-limit        :
-j | --jupyter           :
-d | --notebook-dir      :
-r | --reverse-proxy     :
-D | --dns-domain        :
-s | --sif               :
-B | --bind              :
-nv | --nv               :
-e | --env-modules       :
--conda-env             :
-Q | --quiet
```



carload-spray-koala.expense-user-content.sdsc.edu/lab

File Edit View Run Kernel Tabs Settings Help

examples / Hello_World /

Name	Last Modified
hello_world_cpu.ipynb	7 months ago
hello_world_gpu.ipynb	7 months ago
README.md	7 months ago

Hello World

FileName: hello_world_cpu.ipynb

CPU Version

No package dependencies

```
[8]: print('Hello world!!!!')
Hello world!!!!

[9]: # Import hello module
import hello

# Define a local function
def world2(name):
    print(name)

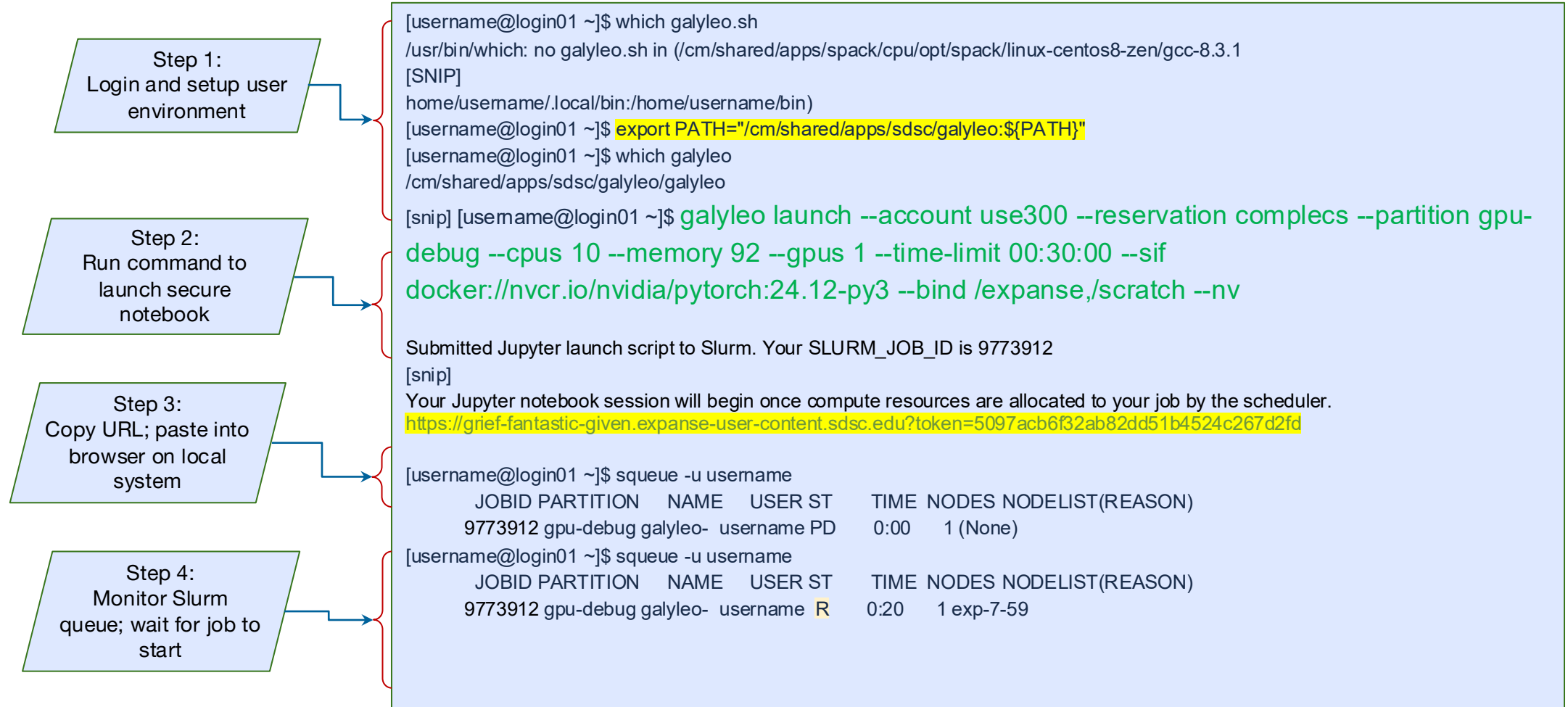
[10]: # Call function
world2("mary")
mary

[11]: hello.greeting("good times")
Greetings, good times

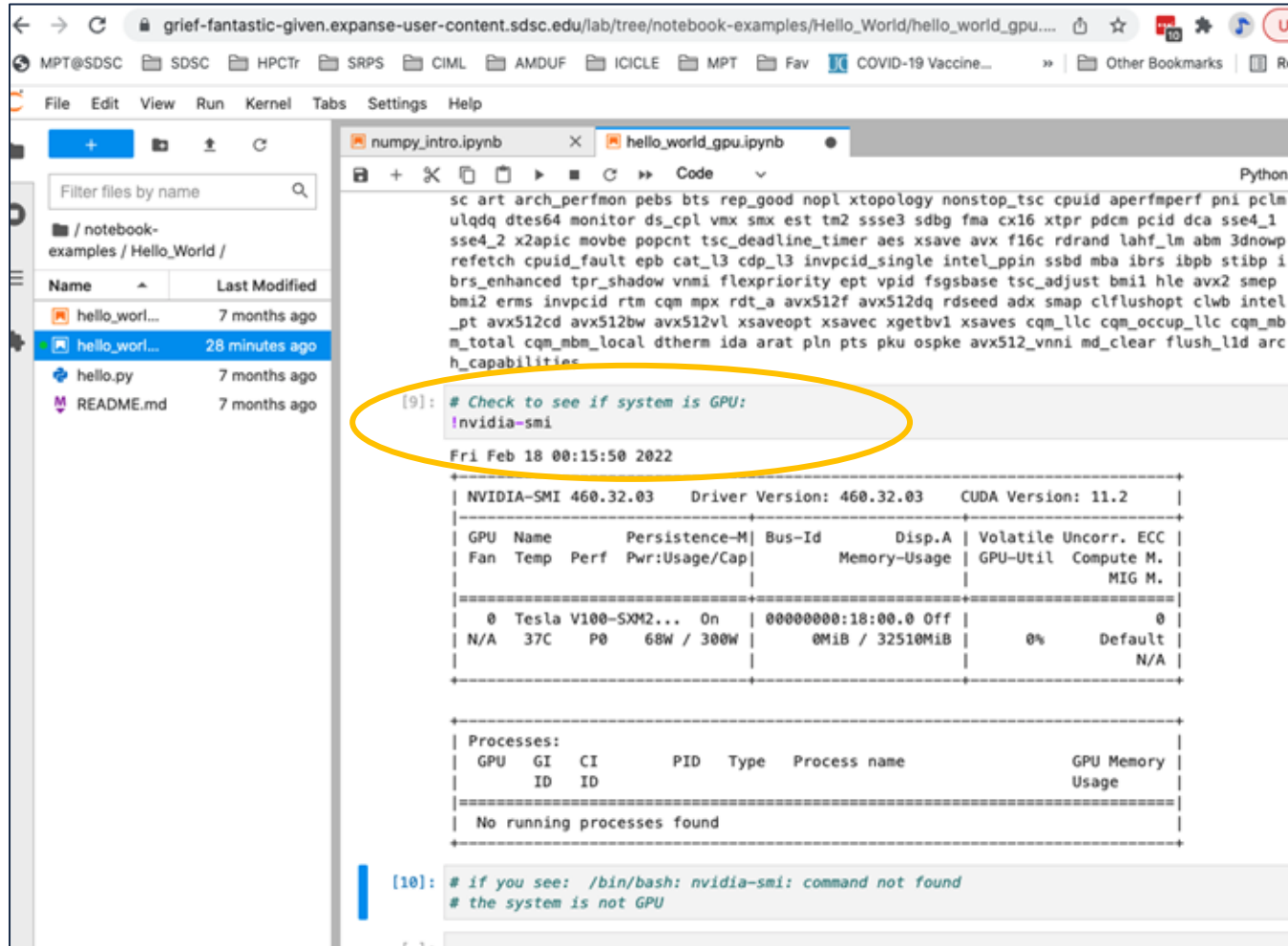
[12]: hello.world("World.")
Hello, World.
```


Launching GPU notebooks using galileo

- GPU Notebooks run better when using containers. SDSC maintains several containers on Expanse
- See: /cm/shared/apps/containers/singularity



Verify notebook launched on GPU device



```
sc art arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfmperf pni pclm
ulqdq dtes64 monitor ds_cpl vmx smx est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid dca sse4_1
sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand lahf_lm abm 3dnowp
refetch cpuid_fault epb cat_l3 cdp_l3 invpcid_single intel_ppin ssbd mba ibrs ibpb stibp i
brs_enhanced tpr_shadow vnmi flexpriority ept vpid fsgsbase tsc_adjust bmi1 hle avx2 smep
bmi2 erms invpcid rtm cqm mpx rdt_a avx512f avx512dq rdseed adx smap clflushopt clwb intel
_pt avx512cd avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves cqm_llc cqm_occup_llc cqm_mb
m_total cqm_mbm_local dtherm ida arat pln pts pku ospke avx512_vnni md_clear flush_lid arc
h_capabilities

[9]: # Check to see if system is GPU:
!nvidia-smi

Fri Feb 18 00:15:50 2022

+-----+
| NVIDIA-SMI 460.32.03   Driver Version: 460.32.03   CUDA Version: 11.2   |
+-----+
| GPU  Name            Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf    Pwr:Usage/Cap|      Memory-Usage | GPU-Util  Compute M. |
|                               |                    |                MIG M. |
+-----+-----+
| 0   Tesla V100-SXM2...  On      | 00000000:18:00:0 Off |   0%      Default |
| N/A   37C    P0     68W / 300W   |  0MiB / 32510MiB |           |
+-----+-----+

+-----+
| Processes: |
| GPU  GI    CI          PID  Type  Process name                      GPU Memory |
|      ID    ID                                   |             Usage |
+-----+-----+
|  No running processes found               |
+-----+

[10]: # if you see: /bin/bash: nvidia-smi: command not found
# the system is not GPU
```

Notebook Examples

<https://github.com/sdsc-hpc-training-org/Expanse-Notebooks>

- Collection of notebooks tested on Expanse and other SDSC HPC systems
- Includes range of materials from “hello world” to ML notebooks.
- View by name, CPU/GPU, or Serial/Parallel

Expanse Notebooks

Refer to this [User Guide](#) for instructions on loading required packages and launching Jupyter Notebook in Expanse.

View by: Name (Alphabetical-Order)

The following table lists the notebooks in alphabetical order. To view by type, use the links below:

- [CPU/GPU](#)
- [Serial/Parallel](#)

Notebook Table: Alphabetical Order

Notebook Project	Notebook	Type	Required
CUDA_GPU_Computing_Pi	cuda_gpu_nvidia_computing_pi_solution.ipynb	GPU, Parallel	numba , mat
CUDA_GPU_Distance_Matrix	cuda_gpu_nvidia_distance_matrix_solution.ipynb	GPU, Parallel	numba , mat
CUDA_GPU_Law_Of_Cosines	cuda_gpu_nvidia_law_of_cosines_solution.ipynb	GPU, Parallel	numba , mat vectorize ,
Clustering_Visualizations	Introduction_to_Clustering.ipynb	CPU, Serial	scikit-learn matplotlib make_blobs dendrogram Agglomerat:
Dask_Graph_CPU	dask_graphs_CPU.ipynb	CPU, Parallel	dask
Dask_Graph_GPU	dask_graphs_GPU.ipynb	GPU, Parallel	dask , cupy array
Data_Analysis	data_analysis_pandas.ipynb	CPU, Serial	numpy , pan
Data_Analysis_Cupy	data_analysis_cupy.ipynb	GPU, Parallel	cupy , cudf numpy
Decision_Trees	Decision trees.ipynb	CPU, Serial	scikit-learn sklearn.dat graphviz ,

Table of Contents: Part 2

- Defining Interactive HPC Computing
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SIMPLIFY! USE
THE EXPANSE
PORTAL

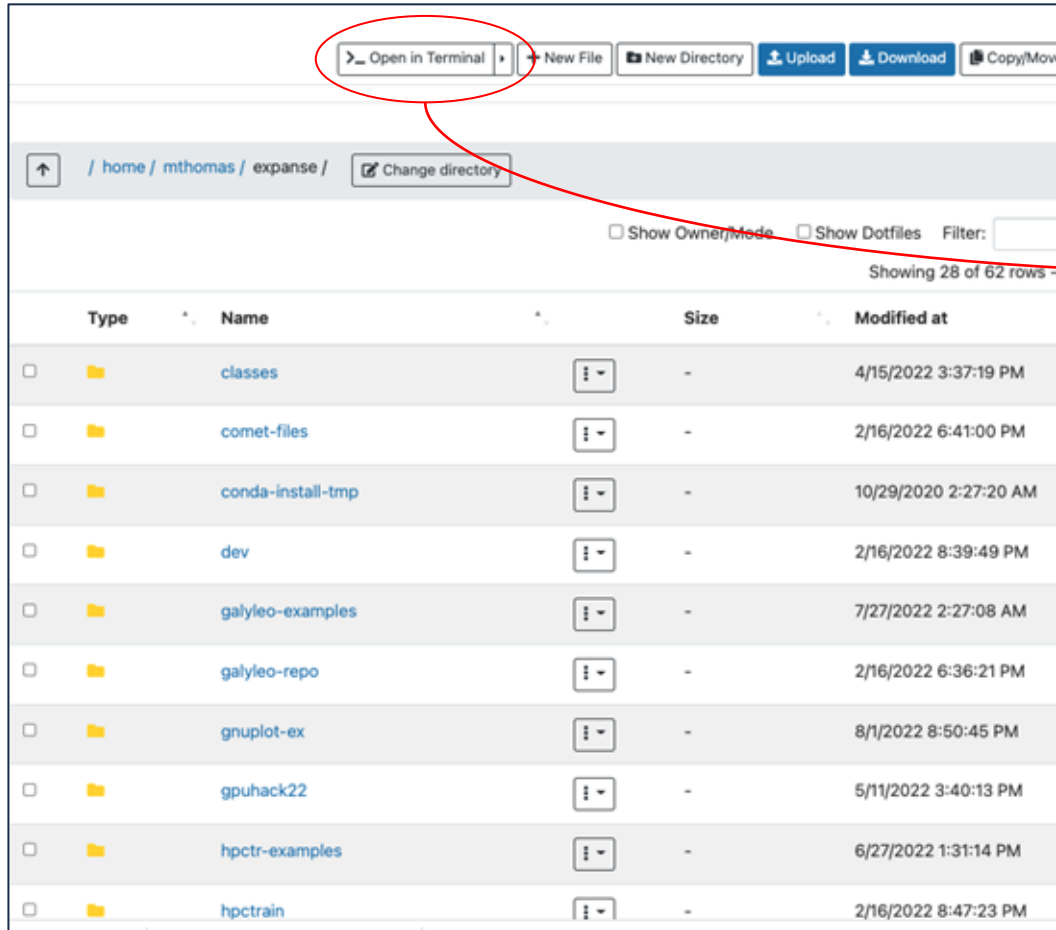
Expanse User Portal

- <https://portal.expense.sdsc.edu>
- authenticate using ACCESS credentials
- Securely hosts batch job submission & monitoring, and interactive applications
- Simplifies launching supported interactive applications
- manages software dependencies
- Based on OOD portal

The collage illustrates the Expanse User Portal's capabilities through several screenshots:

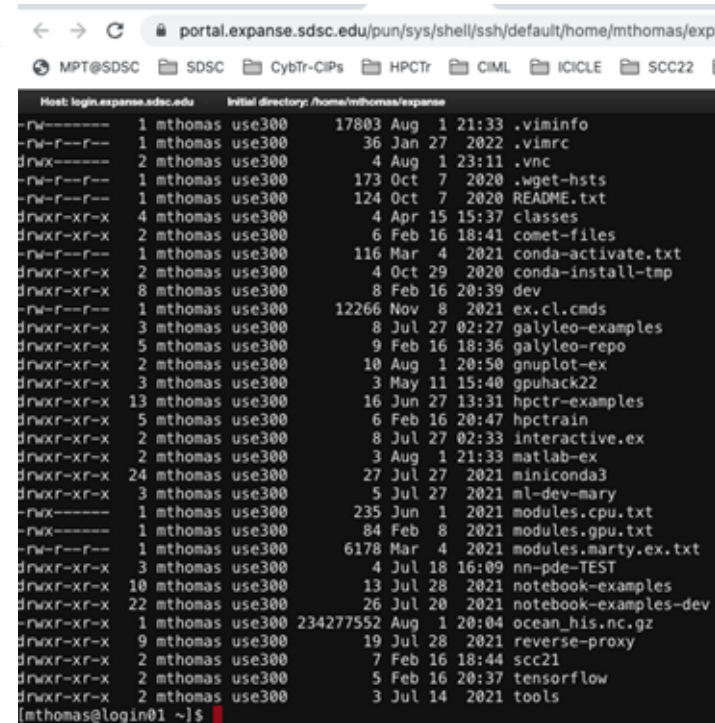
- File Browsing, editing, creation, upload, download, etc.:** A screenshot of the 'Files' section showing a list of files and folders with columns for Name, Size, and Modified at.
- SDSC Expanse Portal:** The main dashboard with a red header and a grid of 'Pinned Apps' including Active Jobs, Home Directory, Job Composer, and others.
- OOD 2.0 Features:** A callout pointing to the top navigation bar and the 'Active Jobs' section.
- Active Jobs:** A detailed view of job status, showing a table with columns for Job ID, Job Name, User, Status, and Time.
- Job Script Editing and Submission:** A screenshot of the 'Jobs' section showing a form for editing and submitting a job script.
- Interactive Services:** A screenshot showing a MATLAB application window with a plot.
- Active Job monitoring and Management:** A screenshot showing a detailed view of a specific job's status and details.

Expanse Portal: File Management



The screenshot shows the Expanse Portal file management interface. The top navigation bar includes buttons for "Open in Terminal", "New File", "New Directory", "Upload", "Download", and "Copy/Move". The "Open in Terminal" button is circled in red, and a red arrow points from it to the terminal window on the right. Below the navigation bar, the breadcrumb path is "/ home / mthomas / expanse /" and there is a "Change directory" button. The main content area displays a table of files and directories. The table has columns for "Type", "Name", "Size", and "Modified at". The table shows 28 of 62 rows.

Type	Name	Size	Modified at
Folder	classes	-	4/15/2022 3:37:19 PM
Folder	comet-files	-	2/16/2022 6:41:00 PM
Folder	conda-install-tmp	-	10/29/2020 2:27:20 AM
Folder	dev	-	2/16/2022 8:39:49 PM
Folder	galileo-examples	-	7/27/2022 2:27:08 AM
Folder	galileo-repo	-	2/16/2022 6:36:21 PM
Folder	gnuplot-ex	-	8/1/2022 8:50:45 PM
Folder	gpuhack22	-	5/11/2022 3:40:13 PM
Folder	hpctr-examples	-	6/27/2022 1:31:14 PM
Folder	hpctrain	-	2/16/2022 8:47:23 PM



```
portal.expanse.sdsc.edu/pun/sys/shell/ssh/default/home/mthomas/expanse
MPT@SDSC SDSC CybTr-CIPs HPCTR CIML ICICLE SCC22 E

Host: login.expanse.sdsc.edu Initial directory: /home/mthomas/expanse
-rw-r--r-- 1 mthomas use300 17803 Aug 1 21:33 .viminfo
-rw-r--r-- 1 mthomas use300 36 Jan 27 2022 .vimrc
drwxr-xr-x 2 mthomas use300 4 Aug 1 23:11 .vnc
-rw-r--r-- 1 mthomas use300 173 Oct 7 2020 .wget-hsts
-rw-r--r-- 1 mthomas use300 124 Oct 7 2020 README.txt
drwxr-xr-x 4 mthomas use300 4 Apr 15 15:37 classes
drwxr-xr-x 2 mthomas use300 6 Feb 16 18:41 comet-files
-rw-r--r-- 1 mthomas use300 116 Mar 4 2021 conda-activate.txt
drwxr-xr-x 2 mthomas use300 4 Oct 29 2020 conda-install-tmp
drwxr-xr-x 8 mthomas use300 8 Feb 16 20:39 dev
-rw-r--r-- 1 mthomas use300 12266 Nov 8 2021 ex.cl.cmds
drwxr-xr-x 3 mthomas use300 8 Jul 27 02:27 galileo-examples
drwxr-xr-x 5 mthomas use300 9 Feb 16 18:36 galileo-repo
drwxr-xr-x 2 mthomas use300 10 Aug 1 20:50 gnuplot-ex
drwxr-xr-x 3 mthomas use300 3 May 11 15:40 gpuhack22
drwxr-xr-x 13 mthomas use300 16 Jun 27 13:31 hpctr-examples
drwxr-xr-x 5 mthomas use300 6 Feb 16 20:47 hpctrain
drwxr-xr-x 2 mthomas use300 8 Jul 27 02:33 interactive.ex
drwxr-xr-x 2 mthomas use300 3 Aug 1 21:33 matlab-ex
drwxr-xr-x 24 mthomas use300 27 Jul 27 2021 miniconda3
drwxr-xr-x 3 mthomas use300 5 Jul 27 2021 ml-dev-mary
-rw-r--r-- 1 mthomas use300 235 Jun 1 2021 modules.cpu.txt
-rw-r--r-- 1 mthomas use300 84 Feb 8 2021 modules.gpu.txt
-rw-r--r-- 1 mthomas use300 6178 Mar 4 2021 modules.marty.ex.txt
drwxr-xr-x 3 mthomas use300 4 Jul 18 16:09 nn-pde-TEST
drwxr-xr-x 10 mthomas use300 13 Jul 28 2021 notebook-examples
drwxr-xr-x 22 mthomas use300 26 Jul 20 2021 notebook-examples-dev
-rw-r--r-- 1 mthomas use300 234277552 Aug 1 20:04 ocean_his.nc.gz
drwxr-xr-x 9 mthomas use300 19 Jul 28 2021 reverse-proxy
drwxr-xr-x 2 mthomas use300 7 Feb 16 18:44 scc21
drwxr-xr-x 2 mthomas use300 5 Feb 16 20:37 tensorflow
drwxr-xr-x 2 mthomas use300 3 Jul 14 2021 tools

[mthomas@login01 ~]$
```

Expanse Portal: Running Matlab

My Interactive Sessions

Interactive Apps

- GUIs
- MATLAB**
- RSTUDIO

MATLAB

This app will launch a MATLAB GUI on Expanse. You will be able to interact with the MATLAB GUI through a VNC session. Please email help@xsede.org to be added to matlab-groups.

Session ID: fee7f0fb-48df-46af-8f4

This app will launch a MATLAB GUI on Expanse. You will be able to interact with the MATLAB GUI through a VNC session. Please email help@xsede.org to be added to matlab-groups.

Partition

compute

Reservation

Number of hours

1

Account

use300

☒ I would like to receive an email when the session starts

Working directory

home

Number of cores

1

Memory (GB)

64

Launch

* The MATLAB session data for this session can be accessed under the data root

portal.expense.sdsc.edu/pun/sys/dashboard/batch_connect/sessions

SDSC CybTr-CIPs HPCTr CIML ICICLE SCC22 MPT AMDUF COVID-19 Vaccine UCSD-SympMon...

My Interactive Sessions

Session was successfully deleted.

My Interactive Sessions

Interactive Apps

- GUIs
- MATLAB**
- RSTUDIO

MATLAB (14834793)

1 node | 128 cores | Running

Host: exp-1-18.expense.sdsc.edu

Created at: 2022-08-01 22:48:55 PDT

Time Remaining:

Session ID:

Compression: 0 (low) to 9 (high)

Launch MATLAB

MATLAB R2022b - academic

HOME PLOTS APPS

File Edit View Insert Tools Desktop Window Help

Current Folder: /home/mthomas

Command Window

```
>> plot(x(1:20))
Error using plot
Invalid first data argument.

>> x = 0:pi/100:2*pi;
>> y = sin(x);
>> plot(x,y)
warning: MATLAB has disabled some advanced rendering features by switching to software rendering. For more information, click here.
```

Figure 1

Plot of a sine wave: $y = \sin(x)$ for $x \in [0, 2\pi]$.

Expanse Portal: Launching Notebooks

The image displays three overlapping screenshots of the Expanse Portal interface, illustrating the process of launching Jupyter notebooks.

Top Left Screenshot: Jupyter Session Configuration

This screenshot shows the 'Jupyter Session' configuration page. The URL is `portal.expanse.sdsc.edu/pun/sys/sdsc_jupyter`. The page includes the following fields:

- Account:** `use300`
- Partition:** `shared` (with a note: "Please choose the gpu, gpu-shared, or gpu-preempt as the partition if using gpus:")
- Time limit (min):** `30`
- Number of cores:** `1`
- Memory required per node (GB):** `2`
- GPUs (optional):** `0`
- Singularity Image File Location:** `/cm/shared/apps/containers/singularity/pytorch/pytorch-latest.sif`

Top Right Screenshot: Jupyter Notebook Environment

This screenshot shows a Jupyter notebook environment. The URL is `blouse-mustang-tusk.expanse-user-content.sdsc.edu/lab/tree/notebook-examples/Hello_W...`. The interface includes a file browser on the left showing a directory structure with files like `hello_worl...` and `hello.py`. The main area displays a code editor with Python code, including a comment: `# Check to see if system is GPU: nvidia-smi`.

Bottom Left Screenshot: Jupyter Session Log

This screenshot shows a 'Jupyter Session' log. The URL is `portal.expanse.sdsc.edu/pun/sys/sdsc_jupyter/ru...`. The log displays session details, including the session ID, time, and a link to the session:

Session ID	Time	Link
2022-08-02 00:13:41	-0700	https://blouse-mustang-tusk.expanse-user-content.sdsc.edu?token=111d82c22973ef486f91a72270e79be
2022-08-02 01:22:26	-0700	https://taste-headcount-rippling.expanse-user-content.sdsc.edu?token=7b4cddde9b116386792d7997a4d7d5c1

Expanse Portal: **Input data** example for Jupyter Notebook

Account: <abc123>

Partition: *(Please choose the gpu, gpu-shared, or gpu-preempt as the partition if using gpus): debug*
Time limit (min): 30

Number of cores: 1

Memory required per node (GB): 2

GPUs (optional): 0

Singularity Image File Location: *(Use your own or to include from existing container library at /cm/shared/apps/container e.g., /cm/shared/apps/containers/singularity/pytorch/pytorch-latest.sif)*
</cm/shared/apps/containers/singularity/pytorch/pytorch-latest.sif>

Environment modules to be loaded *(E.g., to use latest version of system Anaconda3 include cpu,gcc,anaconda3):* singularitypro

Conda Environment *(Enter your own conda environment if any):*

Conda Init *(Provide path to conda initialization scripts):*

Conda Yaml *(Upload a yaml file to build the conda environment at runtime) No file chosen:*

Turn on use of mamba for speeding up conda-yml installs:

Enable use of new caching mechanism that will store and reuse conda-yml created environments using conda-pack !!!!!

Reservation: complecs

QoS:

Working directory: HOME

Type: JupyterLab

Thank You!

Q&A

If you have problems, please contact **consult@sdsc.edu**

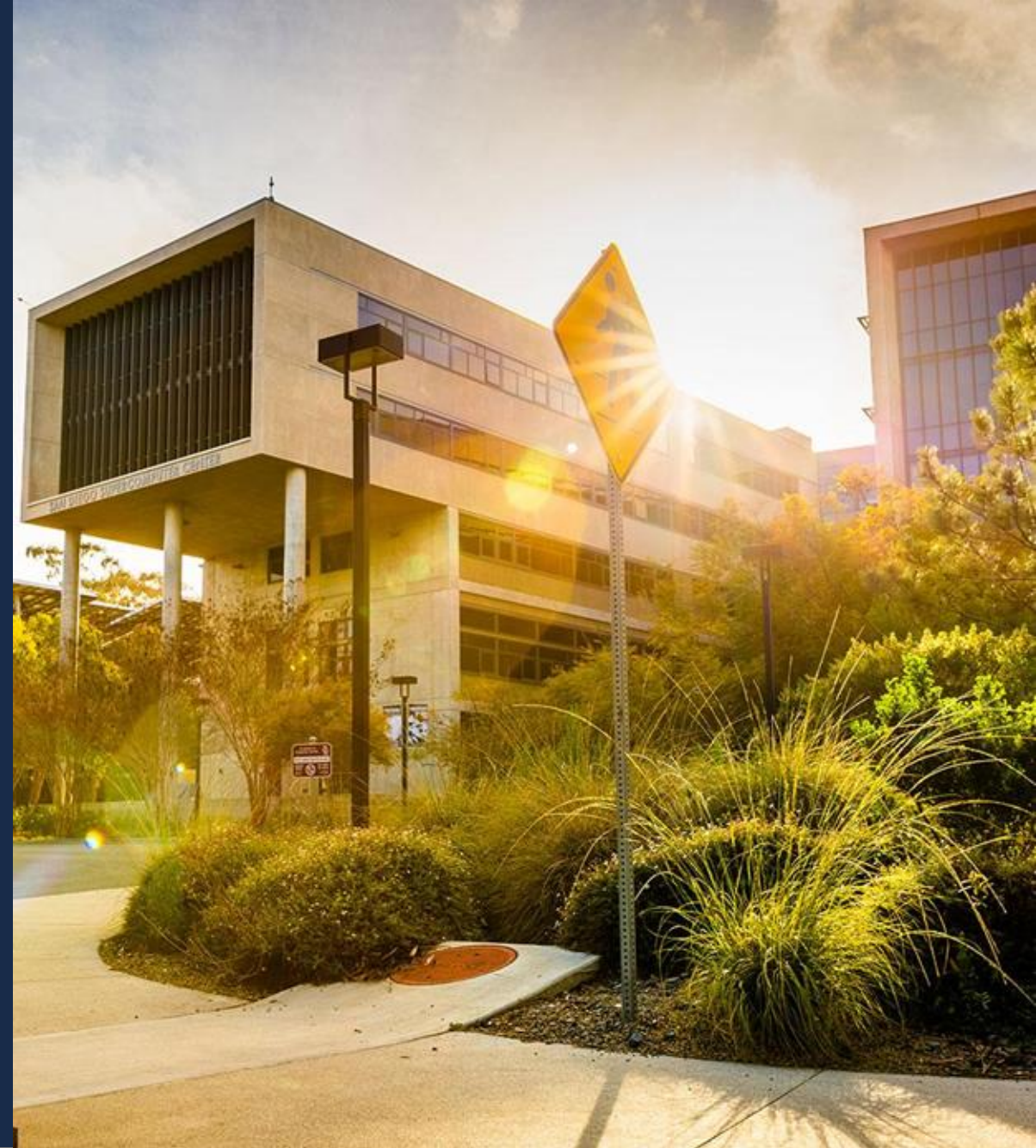
<https://github.com/sdsc-complecs/interactive-computing/>

Some final notes

- When you get started on a new HPC system, or begin working with a new piece of software, you may not know the computational requirements for the types of workloads you need to run. Experiment! Start small on number of CPUs/GPUs, amount of memory, then scale up your job resource requests as needed. Start with long on max wallclock runtimes (`#SBATCH -time`), then dial it back.
- Be aware of how your HPC system charges for the resources requested from the scheduler. Some charge by node-hour, some by CPU-core-hour, some by GPU-hour, some by memory, or even a combination of all of the above. In general, you are only charged for the resources consumed, not those requested.
- Read your HPC system's user guide, the Slurm documentation, and/or the documentation to the specific software you're attempting to run.
- If you have any questions, please don't hesitate to contact your HPC user services support team. We are there to help you understand how to utilize these specialized resources effectively for your research.

Additional Information

- [Slurm Quick Start User Guide](#)
- [HPCWIKI: Scheduling Basics](#)
- [Introduction to HPC](#)
- [HPC in a Day](#)
- [Google's Shell Style Guide](#)



Scheduler Rosetta Stone

Common User Commands	PBS/Torque	Slurm	LSF	SGE	LoadLeveler
Job submission	qsub [script_file]	sbatch [script_file]	bsub [script_file]	qsub [script_file]	lsubmit [script_file]
Job deletion	qdel [job_id]	scancel [job_id]	bkill [job_id]	qdel [job_id]	llcancel [job_id]
Job status	qstat [job_id]	squeue [job_id]	bjobs	qstat [-j job_id]	llq -u [username]
Job status (by user)	qstat -u [user_name]	squeue -u [user_name]	bjobs	qstat [-j job_id]	llq -u [user_name]
Job hold	qhold [job_id]	scontrol hold [job_id]	bstop [job_id]	qhold [job_id]	llhold -r [job_id]
Job release	qrls [job_id]	scontrol release [job_id]	brresume [job_id]	qrls [job_id]	llhold -r [job_id]
Queue list	qstat -Q	squeue	bqueues	qconf -sq	llclass
GUI	xpbsmon	sview	xist/xsbatch	N/A	xload
Environment Variables	PBS/Torque	Slurm	LSF	SGE	LoadLeveler
Job ID	\$PBS_JOBID	\$SLURM_JOBID	\$LSB_JOBID	\$JOB_ID	\$LOAD_STEP_ID
Working Directory	\$PBS_O_WORKDIR	Use "pwd" command	Use "pwd" command	\$SGE_O_WORKDIR	\$LOADL_STEP_INITDIR
Node List	\$PBS_NODEFILE	\$SLURM_JOB_NODELIST	\$LSB_HOSTS/LSB_MCPU_HOST	\$PE_HOSTFILE	\$LOADL_PROCESSOR_LIST
Job Specification	PBS/Torque	Slurm	LSF	SGE	LoadLeveler
Script directive	#PBS	#SBATCH	#BSUB	#\$	#@
Queue	-q [queue]	-p [queue]	-q [queue]	-q [queue]	class=[queue]
Node Count	-l nodes=[count]	-N [min[-max]]	-n [count]	N/A	node=[count]
CPU Count	-l ppn=[count]	-n [count]	-n [count]	-pe ompi [#]	tasks_per_node=[count]
Wall Clock Limit	-l walltime=[hh:mm:ss]	-t [min] OR -t [days-hh:mm:ss]	-W	-l time=[hh:mm:ss]	wall_clock_limit=[hh:mm:ss]
Standard Output File	-o [file_name]	-o [file_name]	-o [file_name]	-o [path]	output=[file]
Standard Error File	-e [file_name]	e [file_name]	-e [file_name]	-e [path]	error=[file]
Combine stdout/err	-j oe (both to stdout) OR -j eo (both to stderr)	(use -o without -e)	(use -o without -e)		
Copy Environment	-V	--export=[ALL NONE variables]	?	-V	environment=COPY_ALL
Event Notification	-m abe	--mail-type=[events]	-B or -N	-m abe	notification=start error complete never always
Email Address	-M [address]	--mail-user=[address]	-u [address]	-M [address]	notify_user=[address]
Job Name	-N [name]	--job-name=[name]	-J [name]	-N [name]	job_name=[name]
Job Restart	-r [y/n]	--requeue OR --no-requeue (NOTE: configurable default)	-r	-r [yes/no]	restart=[yes/no]
Job Type	N/A	N/A	N/A	N/A	job_type=[type]
Working Directory	N/A	--workdir=[dir_name]	(submission directory)	-wd [directory]	initialdir=[directory]
Resource Sharing	-l raccesspolicy=singlejob	--exclusive OR --shared	-x	N/A	node_usage=not_shared
Memory Size	l mem=[MB]	--mem=[MB] OR --mem-per-cpu=[MB]	-M [MB]	-mem [GB]	requirements=(Memory >= [MB])
Project to charge	-W group_list=<project>	--account=[project]	-P <project>		
Resource Requirements			-R		
Job dependency		--depend=[job_id:state]	-w [done exit finish]		
Job project		--wckey=[name]	-P [name]		
Job host preference		--nodelist=[nodes] AND/OR --exclude=[nodes]	-m [nodes]		

<https://slurm.schedmd.com/rosetta.html>

Resources

- GitHub Repo for this presentation:
 - <https://github.com/sdsc-complecs/interactive-computing/>
- SDSC Training Resources
 - https://www.sdsc.edu/education_and_training/training
 - Code: <https://github.com/sdsc-hpc-training-org/hpctr-examples>
 - Running notebooks
 - using *galileo*: <https://github.com/mkandes/galileo>
 - Examples: <https://github.com/sdsc-hpc-training-org/notebook-examples>
- Expanse :
 - Landing page: expance.sdsc.edu
 - User Guide: https://expance.sdsc.edu/support/user_guides/expance.html
 - Training: https://www.sdsc.edu/education_and_training/training_hpc.html
- Problems? Contact consult@sdsc.edu